Linear Algebra

MATH10015 · Group 07 · Linear maps, their three forms, kernel, image and rank–nullity

FRIDAY 6 NOVEMBER 2026 · 09:00–10:00 · FRY BLDG G.14

Linear Algebra — lesson plan (55 min)

Linear maps, their three forms, kernel, image and rank–nullity.

Elapsed	What	Budget
00–04	Logistics (1 min) and the key-ideas page (3 min).	Start L1 at 04:00
04–10	L1 · Definition of a linear map: which of these four are linear? (LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Definition of a linear map; Definition 7.1, Lemma 7.4(i), Example 7.22)	6 min
10–14	L2 · Challenge: one condition holds, the other fails (LA tutor guide Week 7 (R. Carey, 29 Oct 2026): challenge in footnote 8; links to PS6 Q3(iii))	4 min
14–21	L3 · Three forms of a linear map: Q7 and Q8 (rotation by π/2 and its inverse) (LA PS6 Q7(a)–(c), Q8(a)–(f); pair activity from LA tutor guide Week 7 (R. Carey, 29 Oct 2026))	7 min
21–28	L4 · Standard basis form: S and T, and their images and kernels (LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Standard basis form; Lemma 7.9, Remark 7.10)	7 min
28–34	L5 · Image and kernel of two matrices (LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Image and kernel; Fact 7.14, Definition 7.26)	6 min
34–39	L6 · Order of composition: $h \circ g$ versus $g \circ h$ (LA PS6 Q9(e))	5 min
39–45	L7 · Bases of the kernel and the image: PS6 Q5(d) (LA PS6 Q5(d))	6 min
45–52	L8 · Rank–Nullity: the picture, today's examples, and the field (LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Rank–Nullity; Theorem 7.31, Proposition 7.33, Corollary 7.35; PS6 Q4 for Q3(ii))	7 min
52–55	Exit check and reminders.	Finish 55:00

Opening words: “A linear map is the kind of function that respects the vector-space structure. Today we test linearity, move between the three forms of a linear map, and then measure what it kills (the kernel) and what it reaches (the image).”

> 00–01 提醒 PS6 hand-in Q1 下周三（11 月 11 日）中午交，只给方法提示；PS5 已截止。01–04 用知识点页过 Definition 7.1、三种形式、Fact 7.14、ker/Im 与 Theorem 7.31（周四 2–3 点已讲）。主线按课程组第 7 周 tutor guide（Rachael 10 月 29 日邮件）编排，PS6 的 Q7、Q8、Q9(e)、Q5(d)、Q4 穿插其中，其余 PS6 题在备用答案库。

If the room is behind: If behind at minute 27 (end of L3): in L4 compute S and S∘T on the board and only state T and T∘S. If behind at minute 40: do L6 with one matrix product only and in L7 find the kernel and state the image. Do not cut L8 — Rank–Nullity is the new theorem of the week.

Each item: show the full question → say the idea → students write a first step → full boardwork → one-line check. The reserve bank below has answers for every other question on the sheet, for questions from the room.

00–01 · LOGISTICS

Course status: read before starting (tutor only)

Logistics for Friday 6 November (tutor card, not printed).

- **PS6 hand-in:** Question 1 only. Students submit on Blackboard by **noon on Wednesday 11 November** (Week 8) for tutor feedback. In the room give hints only (item R-Q1): no verdicts, no matrices.
- **PS5 hand-in:** closed at noon on Wednesday 4 November; feedback is being prepared.
- **Numbas quiz:** an issue was noted in the 4 November course announcement. If students ask, tell them to follow the announcement and not to worry about how the quiz mark is displayed.
- **Rank–Nullity:** lectures are Mon 1–3, Tue 9–11 and Thu 2–3, so Theorem 7.31 was lectured on Thursday 5 November, before this tutorial. L8 assumes it; a one-line “ask the room” is enough.
- **Plan:** the main route follows the course team's Week 7 tutor guide (definition of a linear map, the forms of a linear map, image and kernel, Rank–Nullity), with PS6 Q7, Q8, Q9(e), Q5(d) and Q4 built in. All other PS6 questions are in the reserve bank (R-…).

Opening script (about 30 seconds).

1. "The PS6 hand-in is Question 1 only, on Blackboard by noon next Wednesday, 11 November. Full sentences: for each part say linear or not and prove it; for the linear ones give the matrix form."

2. "PS5 hand-in closed on Wednesday; feedback is on its way."

3. "If the Numbas quiz looks odd, follow the 4 November announcement and don't worry about the mark display."

4. "Today: what makes a map linear, its three forms, kernels and images, and the Rank–Nullity Theorem from Thursday's lecture."

Route (48 min): L1 definition (6) → L2 one condition but not the other (4) → L3 PS6 Q7+Q8 (7) → L4 standard basis form, S, T (7) → L5 image and kernel of two matrices (6) → L6 PS6 Q9(e) (5) → L7 PS6 Q5(d) (6) → L8 Rank–Nullity (7).

> 开场 1 分钟内说完，不展开。Q1 是交作业题：课上只给方法提示（R-Q1），不说结论、不写矩阵。Numbas 的问题只让学生看 11 月 4 日公告，不评论分数显示。Rank–Nullity 周四（2–3 点）已讲，L8 不再设前提；到 L8 时问一句“周四画过这张图吗？”即可。

TEACHER ONLY · HOW TO READ THE SYMBOLS

Symbols: how to say them / 符号读法

Everything that appears in this week's Linear Algebra questions. Each main-route card also has a "Read aloud" line for its own formulas.

Symbol	Say	中文
$\mathbb{R}^2,\ \mathbb{R}^3,\ \mathbb{C}^4$	"R two, R three, C four"	二维实空间等
$T\colon V \to W$	"T from V to W"	从 V 到 W 的映射
$K\text{-linear},\ \mathbb{R}\text{-linear}$	"K-linear, R-linear"	在 K / R 上线性
$e_1,\ e_2$	"e one, e two"	标准基向量
$T(e_1) = e_2$	"T of e one equals e two"	T 把 e_1 送到 e_2
$\frac{1}{2}e_1$	"a half e one"	二分之一 e_1
$\begin{pmatrix} x \\ y \end{pmatrix}$	"the vector x, y"	列向量
$x \mapsto Ax$	"x maps to A x"	x 映到 Ax
$A \in M_{2,3}(\mathbb{R})$	"A is a real two-by-three matrix"	2×3 实矩阵
$g^{-1},\ A^{-1}$	"g inverse, A inverse"	逆映射、逆矩阵
$h \circ g$	"h composed with g / h after g (do g first)"	复合，先 g 后 h
$A_h A_g$	"A h times A g"	矩阵乘积
$\ker T,\ \operatorname{Im} T$	"the kernel of T; the image of T"	核；像
$\operatorname{rank} T,\ \operatorname{nullity} T$	"the rank of T; the nullity of T"	秩（像的维数）；零化度（核的维数）
$\dim V,\ \dim_{\mathbb{C}} \mathbb{C}^4$	"the dimension of V; the dimension of C four over C"	维数；在 C 上的维数
$\operatorname{span}\{v_1, v_2\}$	"the span of v one and v two"	张成的子空间
$S(A, 0)$	"S of A, zero: the solutions of A x equals zero"	齐次方程组的解集
$u_1 \cdot x$	"u one dot x"	点积
$\overline{z},\ \operatorname{Re}(z)$	"z bar (the conjugate); the real part of z"	共轭；实部
$\lambda,\ \theta$	"lambda, theta"	希腊字母
$\det A$	"the determinant of A / det A"	行列式
$\{0\}$	"the zero subspace / just the zero vector"	零子空间
$y^2,\ \sin x,\ e^{x_1}$	"y squared; sine x; e to the x one"	平方；正弦；指数

START HERE · 01–04 MINUTES

Key ideas: linear maps, kernel, image, rank

Linear Algebra · Week 7 · Chapter 7 · Keep this page beside your working.

1 · What "linear" means

Let V, W be vector spaces over a field K. A function $T : V \to W$ is **K-linear** (Definition 7.1) if for all $v, w \in V$ and $\lambda \in K$:

$$T(v + w) = T(v) + T(w), \qquad T(\lambda v) = \lambda T(v).$$

To prove linear: check both identities for *arbitrary* vectors and scalars. **To prove not linear:** one explicit counterexample is enough. A quick first test: every linear map has $T(0) = 0$ (Lemma 7.4(i)) — but $T(0) = 0$ alone does *not* prove linearity (e.g. $(x, y) \mapsto (0, \sin x)$).

The field matters. A map $\mathbb{C}^n \to \mathbb{C}^m$ can be $\mathbb{R}$-linear without being $\mathbb{C}$-linear: test the scalar $\lambda = i$.

2 · Three forms of the same linear map

Form	Example: $f(x, y, z) = (3x,\ x + y + z)$
Algebraic form	$f(x, y, z) = (3x,\ x + y + z)$
Standard basis form	$f(e_1) = 3e_1 + e_2,\ f(e_2) = e_2,\ f(e_3) = e_2$
Matrix form	$f(x) = Ax,\ A = \begin{pmatrix} 3 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \in M_{2,3}(\mathbb{R})$

The Fact (7.14): Ae_i is the ith column of A. So the columns of the matrix are $f(e_1), \ldots, f(e_n)$, and a linear map is determined by where it sends $e_1, \ldots, e_n$ (Remark 7.10). Size check: $f : \mathbb{R}^n \to \mathbb{R}^m$ has an $m \times n$ matrix. Composition matches multiplication: the matrix of $h \circ g$ is $A_h A_g$ (apply g first).

Warning (Example 7.20). You can always build a matrix from $f(e_1), \ldots, f(e_n)$; it represents f only if it agrees with f on *every* input.

3 · Kernel, image, rank and nullity

$$\ker T = \{v \in V : T(v) = 0\}, \qquad \operatorname{Im} T = \{T(v) : v \in V\}, \qquad \operatorname{nullity} T = \dim \ker T, \quad \operatorname{rank} T = \dim \operatorname{Im} T.$$

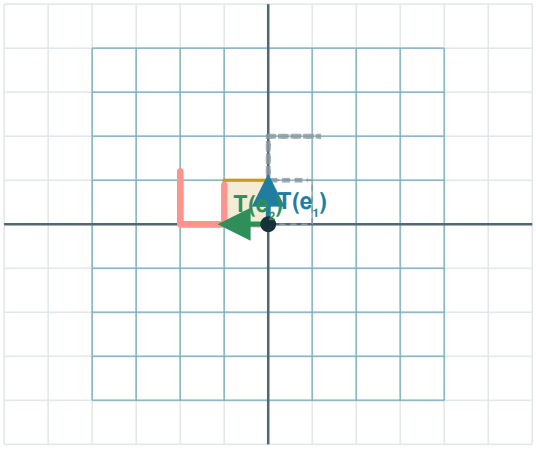
Both are subspaces (Lemma 7.27). T is injective $\iff \ker T = \{0\}$; T is surjective $\iff \operatorname{Im} T = W$. For $T(x) = Ax$, $\operatorname{Im} T$ is the span of the columns and $\ker T = S(A, 0)$.

Rank–Nullity Theorem (Theorem 7.31). For $T : V \to W$ with $\dim V = n$ finite, $\operatorname{rank} T + \operatorname{nullity} T = n = \dim V$. Count dimensions over the stated field: $\dim_{\mathbb{C}} \mathbb{C}^4 = 4$ but $\dim_{\mathbb{R}} \mathbb{C}^4 = 8$.

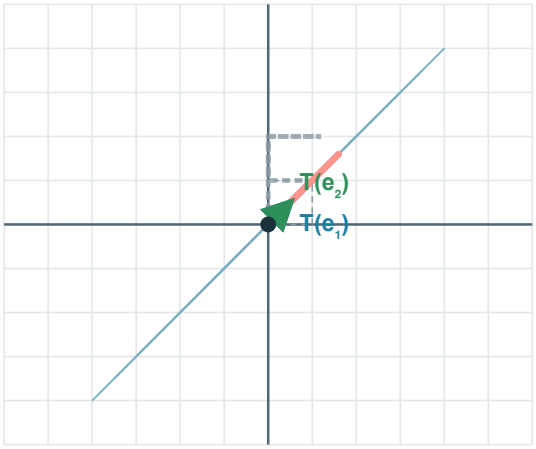
VISUAL · USE DURING L1, L3–L4 AND L6

What a linear map does to the grid

Dashed: the unit square and the letter F before the map. Solid: their images. The blue and green arrows are $T(e_1)$, $T(e_2)$ — the columns of the matrix (Fact 7.14).



Matrix columns = images of e_1, e_2: (0, 1) and (-1, 0)



Matrix columns = images of e_1, e_2: (0.5, 0.5) and (0.5, 0.5)

Q8 g: rotation π/2: Columns are $g(e_1) = e_2$ and $g(e_2) = -e_1$.

Main route · 04–52

8 items, 48 minutes including student writing time.

MAIN ROUTE · 04–10 elapsed · 6 min

L1 · Definition of a linear map: which of these four are linear?

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Definition of a linear map; Definition 7.1, Lemma 7.4(i), Example 7.22

(a) State the definition of an $\mathbb{R}$-linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$. How will you remember it?

(b) Decide whether each of the following maps $\mathbb{R}^2 \to \mathbb{R}^2$ is $\mathbb{R}$-linear:

$$T_1 : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+2 \\ y+2 \end{pmatrix}, \quad T_2 : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x \\ 3y \end{pmatrix}, \quad T_3 : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+y \\ x-y \end{pmatrix}, \quad T_4 : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+y \\ y^2 \end{pmatrix}.$$

(c) What do the linear ones have in common, and what goes wrong in the others? Suggest a general form for a linear map $\mathbb{R}^2 \to \mathbb{R}^2$.

(d) Lemma 7.4(i) says that a linear map satisfies $T(0) = 0$. Show that $g : \mathbb{R}^2 \to \mathbb{R}^2$, $g(x, y) = (0, \sin x)$, satisfies $g(0) = 0$ but is not linear.

> **中文题意：**写出线性映射的定义，判断四个映射是否线性。关键：线性映射的每个分量都是 ax+by，不带常数项；T(0)=0 只是必要条件，(0, sin x) 满足它但不是线性。

> **Read aloud / 读法**
> - $T : \mathbb{R}^2 \to \mathbb{R}^2$ — “T from R two to R two”
> - $\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+y \\ y^2 \end{pmatrix}$ — “the vector x, y maps to x plus y, y squared”
> - $g(x, y) = (0, \sin x)$ — “g of x, y equals zero, sine x”

> **Say:** Linear means two things, for all inputs: adding then mapping is the same as mapping then adding, and the same for scaling. Sort these four into linear and not linear, then tell me what the linear ones have in common.

> 板书先写 Definition 7.1 两条，然后给 1 分钟让学生两人一组判断四个映射。引导学生自己说出“每个分量都是 $ax + by$，没有常数项”，再写成矩阵形式 $x \mapsto Ax$。常见错误：T_4 只说“有平方”不给具体数字；看到 $T(0) = 0$ 就以为线性（T_4 和 (d) 的 g 都是反例）。时间紧时 T_3 的证明只写加法一条，(d) 只写数字反例。

Students write a first step (30–60 s) → then board.

(a) (Definition 7.1) T is $\mathbb{R}$-linear if $T(v + w) = T(v) + T(w)$ for all $v, w \in \mathbb{R}^2$, and $T(\lambda v) = \lambda T(v)$ for all $v \in \mathbb{R}^2$, $\lambda \in \mathbb{R}$. To remember it: “add then map = map then add” and “scale then map = map then scale”.

To prove linearity, check both conditions *in general*. To disprove it, one explicit example of one condition failing is enough.

(b) Write $v = (x, y)$, $w = (x', y')$, $\lambda \in \mathbb{R}$.

- T_1: **not linear**. $T_1(0, 0) = (2, 2) \neq (0, 0)$, impossible for a linear map by Lemma 7.4(i). Explicitly, $T_1(0 + 0) = (2, 2)$ but $T_1(0) + T_1(0) = (4, 4)$.

- T_2: **linear**. $T_2(v+w) = \big(2(x+x'),\ 3(y+y')\big) = (2x,3y)+(2x',3y') = T_2(v)+T_2(w)$ and $T_2(\lambda v) = (2\lambda x,\ 3\lambda y) = \lambda(2x,3y) = \lambda T_2(v)$.
- T_3: **linear**. $T_3(v+w) = \big((x+x')+(y+y'),\ (x+x')-(y+y')\big) = (x+y,\ x-y)+(x'+y',\ x'-y') = T_3(v)+T_3(w)$ and $T_3(\lambda v) = (\lambda x+\lambda y,\ \lambda x-\lambda y) = \lambda T_3(v)$.
- T_4: **not linear**. Here $T_4(0)=0$, so the zero test does not help. Take $v=(0,1)$, $\lambda=2$: $T_4(0,2)=(2,4)$ but $2T_4(0,1)=2(1,1)=(2,2)$.

(c) In T_2 and T_3 every component is a *linear expression $ax+by$ with zero constant term*: no constants, no powers, no products. T_1 fails because of the constant $+2$ (so $T_1(0)\neq 0$); T_4 fails because of y^2. General form:

$$T(x,y) = (ax+by,\ cx+dy) \quad \text{i.e.} \quad T(x) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} x,$$

which is linear by Fact 7.14(i). For example T_2 has matrix $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and T_3 has $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Conversely every linear map $\mathbb{R}^2 \to \mathbb{R}^2$ has this form, with columns $T(e_1), T(e_2)$ (Lemma 7.9, Fact 7.14(ii)).

(d) $g(0,0) = (0,\sin 0) = (0,0)$. Take $v=(\frac{\pi}{2},0)$, $\lambda=2$: $g(2v) = g(\pi,0) = (0,\sin\pi) = (0,0)$, but $2g(v) = 2\big(0,\sin\frac{\pi}{2}\big) = (0,2)$. So g is not linear. (Example 7.22: the matrix agreeing with g on e_1, e_2 gives $(0, x\sin 1)$, which is not $\sin x$.) *Moral: $T(0)=0$ is necessary for linearity but not sufficient.*

> **Check:** Is $(x,y)\mapsto(3x-y,\ 0)$ linear? Is $(x,y)\mapsto(xy,\ x)$?
> **Answer:** The first is linear (matrix $\begin{pmatrix} 3 & -1 \\ 0 & 0 \end{pmatrix}$). The second is not: $(2,2) = 2(1,1)$ is sent to $(4,2)$, but $2\cdot(1,1) = (2,2)$.

MAIN ROUTE · 10–14 elapsed · 4 min

L2 · Challenge: one condition holds, the other fails

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): challenge in footnote 8; links to PS6 Q3(iii)

A map can satisfy one condition of Definition 7.1 but not the other.

(a) Let $f : \mathbb{C} \to \mathbb{C}$, $f(z) = \overline{z}$ (complex conjugation). Show that $f(z + w) = f(z) + f(w)$ for all $z, w \in \mathbb{C}$, but that $f(\lambda z) = \lambda f(z)$ fails for some $\lambda \in \mathbb{C}$. Is f $\mathbb{R}$-linear?

(b) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x, y) = \begin{cases} x & \text{if } y = 0, \\ y & \text{if } y \neq 0. \end{cases}$$

Show that $f(\lambda v) = \lambda f(v)$ for all $\lambda \in \mathbb{R}$ and $v \in \mathbb{R}^2$, but that f is not additive.

中文题意：找只满足线性定义中一个条件的映射。关键：复共轭满足加法但对 λ = i 不满足数乘（它是 R-线性、不是 C-线性）；分段函数满足数乘但 f(1,1)=1 ≠ 2。

Read aloud / 读法

- $f(z) = \overline{z}$ — “f of z equals z bar” (the complex conjugate of z)
- $f(\lambda z) = \lambda f(z)$ — “f of lambda z equals lambda times f of z”
- $\lambda \in \mathbb{C}$ — “lambda in C”

Say: Here is a challenge: can a map pass one test of linearity and fail the other? It can, both ways round, and these two examples show it.

先给学生 30 秒自己想例子，再给出两个。常见错误：(a) 用 $\lambda = i$ 的反例说“不是 $\mathbb{R}$-linear”，但 i 不是实数标量；(b) 忘了分 $\lambda = 0$、$y = 0$ 的情况。连到 PS6 Q3(iii)：共轭和 Re 一样，是 $\mathbb{R}$-linear 而不是 $\mathbb{C}$-linear。时间紧时 (b) 的齐次性只讲 $y \neq 0$ 一种情况，其余口头补。

Students write a first step (30–60 s) → then board.

(a) Additive. Write $z = a + bi$, $w = c + di$ with $a, b, c, d \in \mathbb{R}$. Then

$$f(z + w) = \overline{(a + c) + (b + d)i} = (a + c) - (b + d)i = (a - bi) + (c - di) = f(z) + f(w).$$

Not $\mathbb{C}$-homogeneous. Take $z = 1$, $\lambda = i$: $f(i \cdot 1) = \overline{i} = -i$, but $i\, f(1) = i \cdot 1 = i$. So f is not $\mathbb{C}$-linear.

But $\mathbb{R}$-linear. For real λ, $f(\lambda(a + bi)) = \overline{\lambda a + \lambda bi} = \lambda a - \lambda bi = \lambda f(a + bi)$. So f is $\mathbb{R}$-linear: viewing $\mathbb{C}$ as $\mathbb{R}^2$, it is $(a, b) \mapsto (a, -b)$, the reflection in the real axis, with matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. This is the same phenomenon as PS6 Q3(iii): the only failure is the complex scalar i.

(b) Homogeneous. Let $v = (x, y)$ and $\lambda \in \mathbb{R}$.

- If $\lambda = 0$: $f(0, 0) = 0$ (the case $y = 0$ gives $x = 0$), and $0 \cdot f(v) = 0$.
- If $\lambda \neq 0$ and $y = 0$: then $\lambda y = 0$, so $f(\lambda x, 0) = \lambda x = \lambda f(x, 0)$.
- If $\lambda \neq 0$ and $y \neq 0$: then $\lambda y \neq 0$, so $f(\lambda x, \lambda y) = \lambda y = \lambda f(x, y)$.

Not additive. $f(1,0) = 1$ and $f(0,1) = 1$, but $f\big((1,0) + (0,1)\big) = f(1,1) = 1 \neq 2 = f(1,0) + f(0,1)$.

Moral. Neither condition implies the other, so a proof of linearity must check both.

> **Check:** Does the map f in (b) satisfy $f(0) = 0$? What does that show?
> **Answer:** Yes, $f(0,0) = 0$, and yet f is not linear: one more example that $T(0) = 0$ is not enough.

MAIN ROUTE · 14–21 elapsed · 7 min

L3 · Three forms of a linear map: Q7 and Q8 (rotation by π/2 and its inverse)

LA PS6 Q7(a)–(c), Q8(a)–(f); pair activity from LA tutor guide Week 7 (R. Carey, 29 Oct 2026)

Q7. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = (-x, -y)$ for all $x, y \in \mathbb{R}$.
(a) Write f in standard basis form.
(b) Write f in matrix form.
(c) Using the matrix form, check that f is indeed an $\mathbb{R}$-linear map.

Q8. Let $g : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map with $g(e_1) = e_2$ and $g(e_2) = -e_1$.
(a) Does the information above enable us to find, uniquely, $g(x)$ for all $x \in \mathbb{R}^2$? Justify your answer.
(b) Write g in algebraic form.
(c) Write g in matrix form.
(d) What transformation of $\mathbb{R}^2$ does g represent?
(e) Using the standard basis form, together with linearity, find g^{-1}.
(f) Confirm your answer by finding g^{-1} by: (i) first finding the inverse of the 2×2 matrix from (c); and then (ii) writing this linear map in standard basis form.

中文题意：把 f(x,y)=(−x,−y) 和 g（$e_1 \to e_2$，$e_2 \to -e_1$）写成三种形式，并求 g 的逆。关键：矩阵的列就是 e_1、e_2 的像（Fact 7.14）；g 是逆时针转 π/2，g^{-1} 转回去。

Read aloud / 读法

- $g(e_1) = e_2$ — “g of e one equals e two”
- g^{-1} — “g inverse”
- 2×2 — “two by two”

Say: There are three ways to write the same linear map: a formula in x and y, where e1 and e2 go, or a matrix. The key link is Fact 7.14(ii): the columns of the matrix are exactly where e1 and e2 go. For the inverse, just ask what has to go where to undo g.

板书先画三种形式的三角关系图，标上“第 i 列 $= e_i$ 的像”。导师手册建议的配对活动：一人用一种形式写一个线性映射，同伴把它转换成另外两种形式；有 1 分钟就做一轮，否则留作课后练习。常见错误：把 $g(e_1)$ 写成矩阵的第一行；(e) 中忘了用线性从 $g^{-1}(-e_1) = e_2$ 推出 $g^{-1}(e_1) = -e_2$。追问：$g \circ g$ 是什么？（就是 Q7 的 f）。时间紧时 Q7 只写 (a)(b)，Q7(c) 和 Q8(f)(ii) 口头说。

Students write a first step (30–60 s) → then board.

The three forms: algebraic (a formula, Definition 7.5), standard basis (images of e_1, e_2, Remark 7.12), matrix ($x \mapsto Ax$, Remark 7.15). Link: the ith column of A is the image of e_i (Fact 7.14(ii)).

Q7(a) $f(e_1) = f(1, 0) = (-1, 0) = -e_1$ and $f(e_2) = f(0, 1) = (0, -1) = -e_2$. So $f : e_1 \mapsto -e_1,\ e_2 \mapsto -e_2$.

Q7(b) Columns from (a): $f(x) = A_f x$ with $A_f = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

Q7(c) For any $(a, b) \in \mathbb{R}^2$, $A_f \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -a \\ -b \end{pmatrix} = f(a, b)$. So f agrees with $x \mapsto A_f x$ on every input, and multiplication by a matrix is linear (Fact 7.14(i)); hence f is $\mathbb{R}$-linear.

Q8(a) Yes. Any $x = (a, b)$ is $ae_1 + be_2$, and since g is linear (Lemma 7.4(iii)),

$$g(ae_1 + be_2) = a\,g(e_1) + b\,g(e_2) = ae_2 - be_1 = (-b, a).$$

So $g(x)$ is determined for every x (Lemma 7.9, Remark 7.10).

Q8(b) $g(x, y) = (-y, x)$ for all $x, y \in \mathbb{R}$.

Q8(c) Columns $g(e_1) = (0, 1)$ and $g(e_2) = (-1, 0)$: $A_g = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Q8(d) East e_1 goes to north e_2, north goes to west $-e_1$: g is the anticlockwise rotation by $\pi/2$ about the origin.

Q8(e) $g(e_1) = e_2$ gives $g^{-1}(e_2) = e_1$; $g(e_2) = -e_1$ gives $g^{-1}(-e_1) = e_2$. By linearity $g^{-1}(-e_1) = -g^{-1}(e_1)$, so $g^{-1}(e_1) = -e_2$. Hence

$$g^{-1} : e_1 \mapsto -e_2, \quad e_2 \mapsto e_1, \qquad g^{-1}(x, y) = x(-e_2) + ye_1 = (y, -x),$$

the clockwise rotation by $\pi/2$. Check: $g(g^{-1}(x, y)) = g(y, -x) = (x, y)$.

Q8(f)(i) $\det A_g = 0 \cdot 0 - (-1) \cdot 1 = 1$, so

$$A_g^{-1} = \frac{1}{1}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and multiplying out confirms $A_g A_g^{-1} = I$.

Q8(f)(ii) Columns of A_g^{-1}: $e_1 \mapsto (0, -1) = -e_2$ and $e_2 \mapsto (1, 0) = e_1$, as in (e).

> **Check:** Compute $g(g(x, y))$. Which earlier map is this, geometrically?
> **Answer:** $g(g(x, y)) = g(-y, x) = (-x, -y) = f(x, y)$: two quarter-turns make a half-turn (rotation by π), and indeed $A_g^2 = -I = A_f$.

MAIN ROUTE · 21–28 elapsed · 7 min

L4 · Standard basis form: S and T, and their images and kernels

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Standard basis form; Lemma 7.9, Remark 7.10

Let e_1, e_2 be the standard basis of $\mathbb{R}^2$.

(a) We are told that $S(e_1) = \frac{1}{2}e_1 + \frac{1}{2}e_2 = S(e_2)$. Is this sufficient to define a linear map $S : \mathbb{R}^2 \to \mathbb{R}^2$? How about $T(e_1) = e_2 = T(e_2)$?

(b) Let

$$T:\ e_1 \mapsto e_2,\ \ e_2 \mapsto e_2, \qquad S:\ e_1 \mapsto \tfrac{1}{2}e_1 + \tfrac{1}{2}e_2,\ \ e_2 \mapsto \tfrac{1}{2}e_1 + \tfrac{1}{2}e_2.$$

Find $\mathrm{Im}(X)$ and $\ker(X)$ where $X = S,\ T,\ S \circ T,\ T \circ S$.

中文题意：只给 e_1、e_2 的像能否确定线性映射（能）；求 S、T、S∘T、T∘S 的像和核。关键：先写成代数式 S(x,y) = (x+y)/2 · (1,1)，T(x,y) = (0, x+y)；结果 S∘T = S、T∘S = T，四个的核都是 span{(1,−1)}。

Read aloud / 读法

- $S(e_1) = \frac{1}{2}e_1 + \frac{1}{2}e_2$ — “S of e one equals a half e one plus a half e two”
- $S \circ T$ — “S composed with T” (do T first)
- $\mathrm{Im}(X),\ \ker(X)$ — “the image of X”, “the kernel of X”

Say: A linear map is pinned down by where e1 and e2 go, and there is no rule that they must go to different places. Write each map in algebraic form first; then the kernel and image are easy to read off.

板书先把 S, T 写成代数式：$S(x,y) = \frac{x+y}{2}(1,1)$，$T(x,y) = (0, x+y)$。常见错误：认为 $S(e_1) = S(e_2)$ “不能定义线性映射”；把 $S \circ T$ 的矩阵写成 $A_T A_S$（顺序反）。要点：$S \circ T = S$、$T \circ S = T$（矩阵乘积验证过），因为 S, T 都只依赖 $x + y$ 且输出的坐标和仍是 $x + y$；四个映射的核都是 $\mathrm{span}\{(1,-1)\}$。时间紧时只算 $A_S A_T$，$A_T A_S$ 口头说结果。

Students write a first step (30–60 s) → then board.

(a) Yes, both times. A linear map on $\mathbb{R}^2$ is uniquely determined by the images of e_1, e_2, and *any* choice of images gives exactly one linear map (Lemma 7.9, Remark 7.10). Nothing requires $S(e_1)$ and $S(e_2)$ to be different, or independent. Explicitly, by linearity:

$$S(x,y) = xS(e_1) + yS(e_2) = (x+y)\left(\tfrac{1}{2}, \tfrac{1}{2}\right) = \left(\tfrac{x+y}{2},\ \tfrac{x+y}{2}\right), \qquad T(x,y) = xe_2 + ye_2 = (0,\ x+y).$$

Matrices (columns = images of e_1, e_2): $A_S = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ and $A_T = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.

(b) *Compositions.* The matrix of $S \circ T$ is $A_S A_T$ (apply T first, so A_T is on the right):

$$A_S A_T = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = A_S, \qquad A_T A_S = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = A_T.$$

So $S \circ T = S$ and $T \circ S = T$, as maps and as matrices. Direct check: $S(T(x,y)) = S(0, x+y) = \left(\frac{x+y}{2}, \frac{x+y}{2}\right) = S(x,y)$ and $T(S(x,y)) = T\left(\frac{x+y}{2}, \frac{x+y}{2}\right) = (0, x+y) = T(x,y)$. Reason: both S and T depend only on $x + y$, and both outputs again have coordinate sum $x + y$. Note $S \circ T = S \neq T = T \circ S$: the order

matters.

Kernels. $S(x,y)=0 \iff x+y=0$, and $T(x,y)=0 \iff x+y=0$. So

$$\ker S = \ker T = \ker(S\circ T) = \ker(T\circ S) = \{(t,-t) : t\in\mathbb{R}\} = \operatorname{span}\{(1,-1)\}.$$

Images. Every output of S is $\frac{x+y}{2}(1,1)$, and $S(c,c)=(c,c)$ for every $c\in\mathbb{R}$. Every output of T is $(x+y)e_2$, and $T(c,0)=(0,c)$. So

$$\operatorname{Im} S = \operatorname{Im}(S\circ T) = \operatorname{span}\{(1,1)\}\ (\text{the line } y=x), \qquad \operatorname{Im} T = \operatorname{Im}(T\circ S) = \operatorname{span}\{e_2\}\ (\text{the } y\text{-axis}).$$

Geometrically, S projects onto the line $y=x$ along $(1,-1)$, and T projects onto the y-axis along $(1,-1)$. Each of the four maps has rank 1 and nullity 1 (used again in L8).

> **Check:** Without multiplying matrices, what is $S\circ S$?
> **Answer:** $S\circ S = S$: S only depends on $x+y$, and $S(x,y)$ again has coordinate sum $x+y$. Indeed $A_S^2 = A_S$ (a projection).

MAIN ROUTE · 28–34 elapsed · 6 min

L5 · Image and kernel of two matrices

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Image and kernel; Fact 7.14, Definition 7.26

Find $\mathrm{Im}(A)$ and $\ker(A)$, i.e. the image and kernel of the map $x \mapsto Ax$, for the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 5 & 4 \end{pmatrix} \qquad \text{and} \qquad A = \begin{pmatrix} 1 & 2 & 6 \\ 5 & 4 & 6 \end{pmatrix}.$$

How is $\ker(A)$ related to the solution set $S(A, 0)$ from LA1a? Explain why the second matrix must have a non-trivial kernel.

中文题意：求两个矩阵的像和核。关键：核就是 Ax = 0 的解集 S(A,0)，像是列向量张成的空间；2×3 矩阵有 3 个未知数、至多 2 个独立方程，核一定非零：span{(2,−4,1)}。

Read aloud / 读法

- $\ker(A) = S(A, 0)$ — “the kernel of A equals S of A, zero” (the solutions of A x equals zero)
- $x \mapsto Ax$ — “x maps to A x”
- $\begin{pmatrix} 1 & 2 & 6 \\ 5 & 4 & 6 \end{pmatrix}$ — “the two-by-three matrix with rows one, two, six and five, four, six”

Say: The kernel of a matrix is just the solution set of Ax = 0, which you met in LA1a as S(A, 0). The image is the span of the columns, by Fact 7.14.

板书先写两行：$\ker(A) = S(A, 0)$，$\mathrm{Im}(A) = \mathrm{span}\{\text{columns}\}$。第二个矩阵做一步行变换 $r_2 \to r_2 - 5r_1$。常见错误：把第二个矩阵的像当成 $\mathbb{R}^3$ 的子空间（它在 $\mathbb{R}^2$ 里）；kernel 只写一个向量而不写成 span 或集合。点出：3 个未知数、至多 2 个独立方程，必有自由变量，所以核非平凡。时间紧时第一个矩阵用 $\det A = -6 \neq 0$ 一句带过。

Students write a first step (30–60 s) → then board.

Two facts. $\ker(A) = \{x : Ax = 0\} = S(A, 0)$, the solution set of the homogeneous system. By Fact 7.14, $Ax = x_1c_1 + \cdots + x_nc_n$, so $\mathrm{Im}(A) = \mathrm{span}\{\text{columns of } A\}$.

First matrix (a map $\mathbb{R}^2 \to \mathbb{R}^2$). Solve $x + 2y = 0$, $5x + 4y = 0$: the first gives $x = -2y$, and then $-10y + 4y = -6y = 0$, so $y = 0 = x$. Hence $\ker(A) = \{0\}$. The columns $(1, 5)$ and $(2, 4)$ are not multiples of each other, so they are linearly independent and span $\mathbb{R}^2$: $\mathrm{Im}(A) = \mathbb{R}^2$. (Consistent with $\det A = 4 - 10 = -6 \neq 0$.)

Second matrix (a map $\mathbb{R}^3 \to \mathbb{R}^2$). Row reduce $Ax = 0$:

$$\begin{pmatrix} 1 & 2 & 6 \\ 5 & 4 & 6 \end{pmatrix} \xrightarrow{r_2 \to r_2 - 5r_1} \begin{pmatrix} 1 & 2 & 6 \\ 0 & -6 & -24 \end{pmatrix}.$$

So $-6y - 24z = 0$, i.e. $y = -4z$, and $x = -2y - 6z = 8z - 6z = 2z$. With $z = t$ free:

$$\ker(A) = S(A, 0) = \{(2t, -4t, t) : t \in \mathbb{R}\} = \mathrm{span}\{(2, -4, 1)\}.$$

(Check: $2 - 8 + 6 = 0$ and $10 - 16 + 6 = 0$.) The first two columns $(1, 5), (2, 4)$ are already independent, so $\mathrm{Im}(A) = \mathbb{R}^2$. The kernel vector records the dependency of the columns: $2(1, 5) - 4(2, 4) + (6, 6) = 0$.

Why the kernel must be non-trivial. $Ax = 0$ has 3 unknowns but only 2 equations, so at most 2 linearly independent equations: the row echelon form has at most 2 leading entries, leaving at least one free variable, and every free variable gives non-zero solutions. Equivalently, the three columns lie in $\mathbb{R}^2$, so they are linearly dependent.

> **Check:** Is the map $x \mapsto Ax$ for the 2×3 matrix injective? Is it surjective?
> **Answer:** Not injective: $A(2, -4, 1) = 0 = A\,0$. Surjective: $\mathrm{Im}(A) = \mathbb{R}^2$.

MAIN ROUTE · 34–39 elapsed · 5 min

L6 · Order of composition: h∘g versus g∘h

LA PS6 Q9(e)

Let $g : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map with $g(e_1) = e_2$ and $g(e_2) = -e_1$ (Q8), so $g(x) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} x$. Let $h : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map $h(x) = Ax$ where $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

(e) By using the matrix forms of the maps, find $h \circ g$ and $g \circ h$ in algebraic form. Are they equal? Describe each one geometrically.

> **中文题意：** 用矩阵乘法求 h∘g 和 g∘h，比较是否相等。关键：h∘g 的矩阵是 A_h A_g（先做 g，所以 A_g 在右边）；h∘g 是关于 x 轴的反射，g∘h 是关于 y 轴的反射，不相等。

> **Read aloud / 读法**
> - $h \circ g$ — “h composed with g” (or “h after g”): apply g first
> - $A_h A_g$ — “A h times A g” (the matrix of h times the matrix of g)

> **Say:** h∘g means: do g first, then h. With matrices, h(g(x)) = A_h(A_g x), so the matrix of h∘g is A_h A_g: the map you apply first sits on the right, next to x.

> 板书先写 $(h \circ g)(x) = A_h(A_g x) = (A_h A_g)x$ 这一行，再算矩阵乘积。常见错误：把 $h \circ g$ 的矩阵写成 $A_g A_h$（顺序反了）；算完不用 e_1 检验。和 L4 呼应：那里 $S \circ T = S \neq T = T \circ S$，同样不交换。Q9(a)–(d) 在备用 R-Q9；时间紧时只算 $A_h A_g$，另一个口头给结果。

Students write a first step (30–60 s) → then board.

First, $h(x, y) = (y, x)$: h swaps the coordinates, which is the reflection in the line $y = x$ (Q9(b)–(d)). And $g(x, y) = (-y, x)$ is the anticlockwise rotation by $\pi/2$.

Order matters: $(h \circ g)(x) = h(g(x)) = A_h(A_g x) = (A_h A_g)x$. The map applied first, g, sits on the right.

$$A_h A_g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \text{so} \quad (h \circ g)(x, y) = (x, -y).$$

$$A_g A_h = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \text{so} \quad (g \circ h)(x, y) = (-x, y).$$

Direct check: $h(g(x, y)) = h(-y, x) = (x, -y)$ and $g(h(x, y)) = g(y, x) = (-x, y)$.

They are not equal: $(h \circ g)(e_1) = e_1$ but $(g \circ h)(e_1) = -e_1$. Composition of maps, like matrix multiplication, is not commutative.

Geometry. Rotate by $\pi/2$, then reflect in $y = x$: the reflection in the x-axis. Reflect first, then rotate: the reflection in the y-axis. Follow e_1: $e_1 \xrightarrow{g} e_2 \xrightarrow{h} e_1$, whereas $e_1 \xrightarrow{h} e_2 \xrightarrow{g} -e_1$.

> **Check:** Which matrix represents “first apply h, then apply g”?
> **Answer:** That map is $g \circ h$, with matrix $A_g A_h = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$: the map applied first goes on the right.

MAIN ROUTE · 39–45 elapsed · 6 min

L7 · Bases of the kernel and the image: PS6 Q5(d)

LA PS6 Q5(d)

Compute the kernel and image of the $\mathbb{R}$-linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x) = (u_1 \cdot x)v_1 + (u_2 \cdot x)v_2$, where $u_1 = (0, 1, -1)$, $u_2 = (1, 1, 0)$ and $v_1 = (1, 1, 1)$, $v_2 = (1, -1, 0)$. Give a basis of $\operatorname{Im} T$ and $\ker T$.

> **中文题意：** 求 T(x) = (u₁·x)v₁ + (u₂·x)v₂ 的核与像，并各给一组基。关键：v₁、v₂ 线性无关，所以 T(x)=0 等价于 u₁·x = 0 且 u₂·x = 0，核 = span{(−1,1,1)}；像 = span{v₁, v₂}；1 + 2 = 3。

> **Read aloud / 读法**
> - $(u_1 \cdot x)v_1$ — "u one dot x, times v one"
> - $\operatorname{Im} T$, $\ker T$ — "the image of T", "the kernel of T"
> - $\operatorname{span}\{v_1, v_2\}$ — "the span of v one and v two"

> **Say:** Kernel: everything that goes to zero, so solve T(x) = 0. Image: every output is a combination of v1 and v2, so the image is at most their span; the work is to show you get all of it.

> 板书先把 $T(x)$ 展开成坐标式 $(x_1 + 2x_2 - x_3,\ -x_1 - x_3,\ x_2 - x_3)$。常见错误：kernel 写成 "0" 或一个向量而不是集合；Im 只证 "$\subseteq \operatorname{span}$"，没有给出 $x = (\lambda_2, 0, -\lambda_1)$ 证明反向包含。追问：T 单射吗？满射吗？时间紧时 v_1, v_2 的线性无关口头说，直接写 kernel 和 image。

Students write a first step (30–60 s) → then board.

Here $u_1 \cdot x = x_2 - x_3$ and $u_2 \cdot x = x_1 + x_2$, so

$$T(x) = (x_2 - x_3)(1, 1, 1) + (x_1 + x_2)(1, -1, 0) = (x_1 + 2x_2 - x_3,\ -x_1 - x_3,\ x_2 - x_3).$$

v_1, v_2 *are linearly independent:* $\lambda_1 v_1 + \lambda_2 v_2 = (\lambda_1 + \lambda_2,\ \lambda_1 - \lambda_2,\ \lambda_1) = 0$ forces $\lambda_1 = 0$ (third entry) and then $\lambda_2 = 0$.

Kernel. By independence, $T(x) = 0$ if and only if $u_1 \cdot x = 0$ and $u_2 \cdot x = 0$, i.e. $x_2 - x_3 = 0$ and $x_1 + x_2 = 0$. Put $x_3 = t$; then $x_2 = t$ and $x_1 = -t$. So

$$\ker T = \{(-t, t, t) : t \in \mathbb{R}\} = \operatorname{span}\{(-1, 1, 1)\},$$

with basis $\{(-1, 1, 1)\}$. (Check: $T(-1, 1, 1) = (-1 + 2 - 1,\ 1 - 1,\ 1 - 1) = (0, 0, 0)$.)

Image. Every $T(x)$ is a linear combination of v_1, v_2, so $\operatorname{Im} T \subseteq \operatorname{span}\{v_1, v_2\}$. Conversely, let $y = \lambda_1 v_1 + \lambda_2 v_2$ and choose $x = (\lambda_2, 0, -\lambda_1)$. Then $u_1 \cdot x = 0 - (-\lambda_1) = \lambda_1$ and $u_2 \cdot x = \lambda_2 + 0 = \lambda_2$, so $T(x) = y$. Hence $\operatorname{Im} T = \operatorname{span}\{v_1, v_2\}$, and since v_1, v_2 are independent, $\{(1, 1, 1), (1, -1, 0)\}$ is a basis of $\operatorname{Im} T$.

So $\operatorname{nullity} T = 1$ and $\operatorname{rank} T = 2$; this is checked against Rank–Nullity in L8.

> **Check:** Using your answer to (d): is T injective? Is T surjective?
> **Answer:** Neither. $T(-1, 1, 1) = 0 = T(0)$, so T is not injective. $\operatorname{Im} T = \operatorname{span}\{v_1, v_2\}$ is a plane, not $\mathbb{R}^3$: e.g. $(0, 0, 1) \notin \operatorname{Im} T$, since $\lambda_1 v_1 + \lambda_2 v_2 = (0, 0, 1)$ forces $\lambda_1 = 1$ and $\lambda_1 + \lambda_2 = \lambda_1 - \lambda_2 = 0$, impossible.

MAIN ROUTE · 45–52 elapsed · 7 min

L8 · Rank–Nullity: the picture, today's examples, and the field

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Rank–Nullity; Theorem 7.31, Proposition 7.33, Corollary 7.35; PS6 Q4 for Q3(ii)

(a) State the Rank–Nullity Theorem. Picture the domain V as a box (1). Inside it, region (2) is the kernel, which T sends to 0; the rest of the box is mapped onto the image, region (3), inside the codomain W. Explain why "(1) has the same size as (2) and (3) together" is a picture of the theorem.

(b) Verify the theorem for: the maps $S, T, S\circ T, T\circ S$ with $S(e_1)=S(e_2)=\frac{1}{2}e_1+\frac{1}{2}e_2$ and $T(e_1)=T(e_2)=e_2$; the matrices $\begin{pmatrix}1&2\\5&4\end{pmatrix}$ and $\begin{pmatrix}1&2&6\\5&4&6\end{pmatrix}$; and the map $T(x)=(u_1\cdot x)v_1+(u_2\cdot x)v_2$ of PS6 Q5(d).

(c) Find a linear map for which (2) = (1). Which maps make (2) as small as possible? What does the size of the codomain W have to do with it?

(d) (PS6 Q4 for Q3(ii).) For $T:\mathbb{C}^4\to\mathbb{C}^2$, $T(x_1,x_2,x_3,x_4)=(x_1+ix_2,\ x_3+ix_4)$, find the rank and nullity first over $\mathbb{C}$ and then over $\mathbb{R}$, and check the theorem both times.

中文题意：讲 Rank–Nullity 定理（定义域维数 = 核的维数 + 像的维数），用今天的例子逐个核对，再看数域的影响。关键：永远对定义域的维数核对；同一个映射在 C 上是 2+2=4，在 R 上是 4+4=8。

Read aloud / 读法

- $\operatorname{rank}T+\operatorname{nullity}T=\dim V$ — "the rank of T plus the nullity of T equals the dimension of V"
- $T:\mathbb{C}^4\to\mathbb{C}^2$ — "T from C four to C two"
- $\dim_{\mathbb{R}}\mathbb{C}^4=8$ — "the dimension of C four over R is eight" (the real dimension)

Say: Rank–Nullity says the dimension of the domain splits into the part that gets killed, the kernel, and the part that survives as the image. Let's check it on everything we have done today, and then see why the count depends on whether we work over C or over R.

一句话问全班："周四讲 Rank–Nullity 时画过这张图吗？"然后板书画两个方框：左框 (1)$=V$，红色 (2)$=\ker T$ 箭头指向 0，其余部分映满右框里的 (3)$=\operatorname{Im}T$，写 $\dim V=\operatorname{nullity}T+\operatorname{rank}T$。常见错误：用 codomain 的维数核对（如把 2×3 矩阵写成 $1+2=2$）；over $\mathbb{R}$ 仍写 nullity 2。追问：右框（W）比左框小会怎样？（核必非平凡，不能单射）。时间紧时 (b) 只核对 2×3 矩阵和 Q5(d)，(d) 只写结论 $2+2=4$ 对 $4+4=8$。

Students write a first step (30–60 s) → then board.

(a) Theorem 7.31. Let V,W be vector spaces over $\mathbb{K}$, $T:V\to W$ linear, and $\dim V<\infty$. Then

$$\operatorname{rank}T+\operatorname{nullity}T=\dim V,\qquad \operatorname{rank}T=\dim\operatorname{Im}T,\quad \operatorname{nullity}T=\dim\ker T.$$

In the picture, (1) is $\dim V$, (2) is $\operatorname{nullity}T$ and (3) is $\operatorname{rank}T$. Why it holds (sketch from the notes): take a basis $u_1,\dots,u_k$ of $\ker T$ (region (2)) and extend it to a basis $u_1,\dots,u_k,v_1,\dots,v_m$ of V. The extra vectors span "the rest of the box", and $T(v_1),\dots,T(v_m)$ is a basis of $\operatorname{Im}T$ (region (3)). So $\dim V=k+m=\operatorname{nullity}T+\operatorname{rank}T$.

(b) Checks (always against the dimension of the *domain*):

- $S, T, S \circ T, T \circ S$: each has kernel $\operatorname{span}\{(1,-1)\}$ and a line as image, so $1+1=2=\dim \mathbb{R}^2$.
- $\begin{pmatrix} 1 & 2 \\ 5 & 4 \end{pmatrix}$: kernel $\{0\}$, image $\mathbb{R}^2$, so $0+2=2=\dim \mathbb{R}^2$.
- $\begin{pmatrix} 1 & 2 & 6 \\ 5 & 4 & 6 \end{pmatrix}$: kernel $\operatorname{span}\{(2,-4,1)\}$, image $\mathbb{R}^2$, so $1+2=3=\dim \mathbb{R}^3$.
- PS6 Q5(d): kernel $\operatorname{span}\{(-1,1,1)\}$, image $\operatorname{span}\{v_1, v_2\}$, so $1+2=3=\dim \mathbb{R}^3$.

(c) (2) = (1) means $\ker T = V$: the *zero map* $T(v)=0$ for all v, with nullity $\dim V$ and rank 0. (2) is as small as possible when $\ker T = \{0\}$, i.e. nullity 0: exactly the *injective* maps (Proposition 7.33(ii)), for example the identity, or $\mathbb{R}^2 \to \mathbb{R}^3$, $(x,y) \mapsto (x,y,0)$. Then $\operatorname{rank} T = \dim V$. The codomain matters because $\operatorname{Im} T \subseteq W$, so $\operatorname{rank} T \leqslant \dim W$ and

$$\operatorname{nullity} T = \dim V - \operatorname{rank} T \geqslant \dim V - \dim W.$$

If $\dim W < \dim V$, the kernel is forced to be non-trivial and T cannot be injective (Corollary 7.35(ii)), as for the 2×3 matrix above. If $\dim W > \dim V$, then $\operatorname{rank} T \leqslant \dim V < \dim W$ and T cannot be surjective (Corollary 7.35(i)).

(d) Over $\mathbb{C}$**.** $T(x)=0 \iff x_1 = -ix_2$ and $x_3 = -ix_4$, so $\ker T = \{(-is, s, -it, t) : s,t \in \mathbb{C}\} = \operatorname{span}_{\mathbb{C}}\{(-i,1,0,0),(0,0,-i,1)\}$, and these two are independent (entries 2 and 4): $\operatorname{nullity}_{\mathbb{C}} T = 2$. Since $T(a,0,b,0)=(a,b)$, $\operatorname{Im} T = \mathbb{C}^2$ and $\operatorname{rank}_{\mathbb{C}} T = 2$. Check: $2+2=4=\dim_{\mathbb{C}} \mathbb{C}^4$.

Over $\mathbb{R}$**.** Same sets, but only real scalars. With $s = s_1 + is_2$, $t = t_1 + it_2$ ($s_j, t_j \in \mathbb{R}$):

$$(-is,\ s,\ -it,\ t) = s_1(-i,1,0,0) + s_2(1,i,0,0) + t_1(0,0,-i,1) + t_2(0,0,1,i),$$

and these four vectors are $\mathbb{R}$-independent (entries 2 and 4 give $s_1 + is_2 = 0 = t_1 + it_2$), so $\operatorname{nullity}_{\mathbb{R}} T = 4$. The image $\mathbb{C}^2$ has $\mathbb{R}$-basis $(1,0),(i,0),(0,1),(0,i)$, so $\operatorname{rank}_{\mathbb{R}} T = 4$. Check: $4+4=8=\dim_{\mathbb{R}} \mathbb{C}^4$.

Moral. Dimension depends on the field: $\dim_{\mathbb{C}} \mathbb{C}^4 = 4$ but $\dim_{\mathbb{R}} \mathbb{C}^4 = 8$ (basis $e_1, ie_1, \ldots, e_4, ie_4$). For instance $(1,i,0,0) = i \cdot (-i,1,0,0)$ is dependent on $(-i,1,0,0)$ over $\mathbb{C}$ but not over $\mathbb{R}$.

> **Check:** A linear map $T : \mathbb{R}^5 \to \mathbb{R}^3$ has rank 3. What is its nullity? Can it be injective?
> **Answer:** $\operatorname{nullity} T = 5 - 3 = 2$. It cannot be injective, since its kernel is not $\{0\}$.

52–55 · EXIT CHECK

Three quick checks before you leave

1. Is $T(x, y) = (x + 1,\ y)$ linear? One-line reason.
2. A linear map $T : \mathbb{R}^5 \to \mathbb{R}^3$ is surjective. What is its nullity?
3. The matrix of T has columns $(1, 2)$ and $(3, 4)$. Write T in standard basis form.

1. No: $T(0,0) = (1,0) \neq 0$, contradicting Lemma 7.4(i).
2. Rank 3, so nullity $= 5 - 3 = 2$ by Rank–Nullity.
3. $T(e_1) = e_1 + 2e_2$, $T(e_2) = 3e_1 + 4e_2$.

> 第 2 题顺带追问：T 能不能单射？（不能，核非平凡）。离开前再说一次 Q1 下周三中午截止、写完整句子。

Reserve answer bank — Linear

For questions from the room or early finishers. Hand-in questions show hints only.

HAND-IN · HINTS ONLY — do not give the full answer

R-Q1 · Hand-in Q1 (due noon Wed 11 Nov): hints only

LA PS6 Q1 (hand-in)

Hand-in question. Submit your solution on Blackboard by noon on Wednesday 11 November (Week 8) for feedback from your tutor. Present all your answers in complete sentences.

Which of the following functions $f : \mathbb{R}^2 \to \mathbb{R}^2$ define an $\mathbb{R}$-linear map? Justify your answer in each case. For those that are linear, find their matrix form.

(a) $f(x, y) = (-y, -x)$ for all $x, y \in \mathbb{R}$.
(b) $f(x, y) = (x + y,\ x \times y)$ for all $x, y \in \mathbb{R}$.
(c) $f(x, y) = (0, 0)$ for all $x, y \in \mathbb{R}$.
(d) $f(x, y) = (0, 0)$ if $(x, y) \in \mathbb{Q}^2$ and $f(x, y) = (x, y)$ otherwise.

> **Say:** Question 1 is your hand-in, so I will only tell you which test to use, not the answers. For each part decide: prove both conditions in general, or find one counterexample with actual numbers; and for any linear one, give the matrix.

> 这是交作业题（11 月 11 日周三中午截止），只给方法提示，不说哪几个是线性的，也不写矩阵。学生问“(b)/(d) 是不是线性”时反问：“你试了哪个条件？代具体数字算两边了吗？”可以提醒：四个小题的 $f(0)$ 都是 0，零测试一个都判断不了。常见失分：只在 e_1, e_2 上验证；反例不写具体数字。

Hints only: this is assessed work. Do not state verdicts or matrices.

For every part. Lemma 7.4(i) does not settle any of (a)–(d), because each of them sends $(0, 0)$ to $(0, 0)$. So every part needs either the full check of Definition 7.1 or one explicit counterexample.

(a) Test both conditions of Definition 7.1 directly, with general vectors $(x, y), (x', y')$ and a general $\lambda \in \mathbb{R}$, exactly as in Q2(a). Each output coordinate is built from a single input coordinate; see where that leads.

(b) The second coordinate is a product. Write out $f(\lambda v)$ and $\lambda f(v)$ (or $f(v + w)$ and $f(v) + f(w)$) in general and compare them. Then decide: either the equality holds for every choice, which you must show in general, or you can pick specific simple vectors (such as e_1, e_2) and numbers where the two sides differ.

(c) Check both conditions of Definition 7.1 directly; with this formula each side takes one line.

(d) The rule depends on whether the point is in $\mathbb{Q}^2$. Think about multiplying a point of $\mathbb{Q}^2$ by an irrational scalar such as $\sqrt{2}$, or adding two points that are not in $\mathbb{Q}^2$ but whose sum is. Each time, use the correct branch of the rule on each side.

A complete answer must contain, for each part:

- a clear verdict in a full sentence ($\mathbb{R}$-linear or not);
- if linear: both conditions verified for *arbitrary* vectors and scalars (not just examples), then the matrix form with its columns found as $f(e_1), f(e_2)$ (Fact 7.14(ii)), ideally confirmed by multiplying the matrix by a general (x, y);

- if not linear: one explicit counterexample with specific numbers, both sides computed and shown to differ, and a statement of which condition fails.

Common pitfalls: checking only on e_1, e_2 (Remark 7.21: a matrix built from e_1, e_2 need not agree with f elsewhere); treating $f(0) = 0$ as a proof of linearity; giving a matrix form for a map that is not linear.

> **Check:** In which of (a)–(d) does Lemma 7.4(i) ($T(0) = 0$) settle the question on its own?
> **Answer:** None: all four maps send $(0, 0)$ to $(0, 0)$, so each part needs the full definition or an explicit counterexample.

RESERVE / LOOKUP

R-Q2 · Linear or not? Proof versus counterexample, then the matrix check

LA PS6 Q2(a), (b), (c), (e)

Determine which of the following are $\mathbb{R}$-linear functions.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 - x_2,\ 5x_1)$.

(b) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 - x_2^2,\ 5x_1)$.

(c) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 - 1,\ 5x_1)$.

(e) $T : \mathbb{R}^2 \to \mathbb{R}$, $T(x_1, x_2) = e^{x_1} - \cos(\pi x_2)$.

For those that are linear, find their matrix form. How could we use this to confirm that those functions were linear?

> **Say:** To show a map is linear you need both conditions for all inputs, so use letters. To show it is not linear, one example with actual numbers is enough. At the end we build the matrix for (a) from T(e1) and T(e2) and check that it agrees with T on every input.

> 备用题（PS6 Q2，非交作业）。板书先写 Definition 7.1 两条，(a) 用一般的 x, y, λ 写完整等式链；(b)(c)(e) 各给一个具体数字反例。常见错误：(e) 中 $T(0) = 0$，零测试判断不了。注意官方解答 (b) 写成 $10x_2$，应为 $10x_1$（题面无误）。Warning 一段（矩阵只在 e_i 上一致）可连到 L1(c)。

We must check, for all $x, y \in \mathbb{R}^2$ and all $\lambda \in \mathbb{R}$: (i) $T(x + y) = T(x) + T(y)$ and (ii) $T(\lambda x) = \lambda T(x)$ (Definition 7.1). To show a map is *not* linear, one explicit counterexample is enough.

(a) Linear. Let $x = (x_1, x_2),\ y = (y_1, y_2) \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$. Then

$$\begin{aligned} T(x + y) &= T(x_1 + y_1,\ x_2 + y_2) \\ &= \big((x_1 + y_1) - (x_2 + y_2),\ 5(x_1 + y_1)\big) \\ &= \big((x_1 - x_2) + (y_1 - y_2),\ 5x_1 + 5y_1\big) \\ &= (x_1 - x_2,\ 5x_1) + (y_1 - y_2,\ 5y_1) = T(x) + T(y), \end{aligned}$$

$$T(\lambda x) = T(\lambda x_1, \lambda x_2) = (\lambda x_1 - \lambda x_2,\ 5\lambda x_1) = \lambda(x_1 - x_2,\ 5x_1) = \lambda T(x).$$

Both conditions hold for all x, y, λ, so T is $\mathbb{R}$-linear.

(b) Not linear. Take $x = e_2 = (0, 1)$ and $\lambda = 2$: $T(2e_2) = T(0, 2) = (0 - 4,\ 0) = (-4, 0)$, but $2T(e_2) = 2(0 - 1,\ 0) = (-2, 0)$. So $T(2e_2) \neq 2T(e_2)$.

(c) Not linear. $T(0, 0) = (-1, 0) \neq (0, 0)$, which is impossible for a linear map by Lemma 7.4(i). Explicitly: $T(0 + 0) = T(0) = (-1, 0)$ but $T(0) + T(0) = (-2, 0)$.

(e) Not linear. Here $T(0, 0) = e^0 - \cos 0 = 0$, so the zero test does not help. Take $x = e_2 = (0, 1)$ and $\lambda = 2$: $T(0, 2) = e^0 - \cos 2\pi = 1 - 1 = 0$, but $2T(0, 1) = 2(e^0 - \cos \pi) = 2(1 + 1) = 4$. So $T(2e_2) \neq 2T(e_2)$.

Matrix form of (a). By Fact 7.14(ii) the columns are $T(e_1) = T(1, 0) = (1, 5)$ and $T(e_2) = T(0, 1) = (-1, 0)$:

$$A = \begin{pmatrix} 1 & -1 \\ 5 & 0 \end{pmatrix}, \qquad A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 - x_2 \\ 5x_1 \end{pmatrix} = T(x_1, x_2) \ \text{ for all } x_1, x_2 \in \mathbb{R}.$$

How this confirms linearity. T agrees with $x \mapsto Ax$ on *every* input, and multiplication by a matrix is linear (Fact 7.14(i), Remark 7.15). Hence T is linear: a second proof.

Warning (Example 7.20, Remark 7.21). The recipe "columns $= T(e_i)$" produces a matrix for any map, but for a non-linear map it agrees with T only on e_1, e_2, not everywhere. For (b): $T(e_1) = (1, 5)$ and $T(e_2) = (-1, 0)$ give the *same* matrix A as in (a), yet $A\begin{pmatrix}0\\2\end{pmatrix} = \begin{pmatrix}-2\\0\end{pmatrix}$ while $T(0, 2) = (-4, 0)$. For (e): $T(e_1) = e - 1$, $T(e_2) = 2$ give the 1×2 matrix $\begin{pmatrix}e-1 & 2\end{pmatrix}$, which sends $(0, 2)$ to 4, while $T(0, 2) = 0$.

> **Check:** The matrix built from $T(e_1), T(e_2)$ for (b) is the same as for (a). Why does that not make (b) linear?
> **Answer:** The matrix only agrees with (b) on e_1, e_2 (Remark 7.21). At $(0, 2)$ it gives $(-2, 0)$ but $T(0, 2) = (-4, 0)$; linearity needs agreement on all inputs.

RESERVE / LOOKUP

R-Q2d · A map built from u_1, u_2: its matrix has columns u_1, u_2

LA PS6 Q2(d)

Determine whether the following is an $\mathbb{R}$-linear function: $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x) = (x \cdot e_1)u_1 + (x \cdot e_2)u_2$, where

$$u_1 = \begin{pmatrix}-1\\3\end{pmatrix}, \quad u_2 = \begin{pmatrix}3\\3\end{pmatrix}, \quad e_1 = \begin{pmatrix}1\\0\end{pmatrix}, \quad e_2 = \begin{pmatrix}0\\1\end{pmatrix}.$$

If it is linear, find its matrix form. How could we use this to confirm that the function is linear?

> **Say:** Dotting with e1 just picks out the first coordinate, so T(x) is x1 times u1 plus x2 times u2. Where do e1 and e2 go? That gives you the columns.

> 先让学生算 $x \cdot e_1 = x_1$，把 T 化成 $x_1u_1 + x_2u_2$，矩阵的列就是 u_1, u_2。常见错误：把 u_1, u_2 当成矩阵的行。可接在 L1(c)"一般形式"之后做。

Linear. The dot product is linear in each slot: $(x + y) \cdot e_j = x \cdot e_j + y \cdot e_j$ and $(\lambda x) \cdot e_j = \lambda(x \cdot e_j)$. So for $x, y \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$:

$$\begin{aligned} T(x + y) &= (x \cdot e_1 + y \cdot e_1)u_1 + (x \cdot e_2 + y \cdot e_2)u_2 \\ &= \big[(x \cdot e_1)u_1 + (x \cdot e_2)u_2\big] + \big[(y \cdot e_1)u_1 + (y \cdot e_2)u_2\big] = T(x) + T(y), \\ T(\lambda x) &= \lambda(x \cdot e_1)u_1 + \lambda(x \cdot e_2)u_2 = \lambda T(x). \end{aligned}$$

Matrix form. For $x = (x_1, x_2)$ we have $x \cdot e_1 = x_1$ and $x \cdot e_2 = x_2$, so $T(x) = x_1u_1 + x_2u_2$. In particular $T(e_1) = u_1$ and $T(e_2) = u_2$, so by Fact 7.14(ii) the matrix has columns u_1, u_2:

$$A = \begin{pmatrix}-1 & 3\\3 & 3\end{pmatrix}, \qquad A\begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{pmatrix}-x_1 + 3x_2\\3x_1 + 3x_2\end{pmatrix} = x_1\begin{pmatrix}-1\\3\end{pmatrix} + x_2\begin{pmatrix}3\\3\end{pmatrix} = T(x).$$

Confirmation. $T(x) = Ax$ for every $x \in \mathbb{R}^2$, and $x \mapsto Ax$ is linear (Fact 7.14(i)), so T is linear. General lesson: $x \mapsto x_1c_1 + \cdots + x_nc_n$ is the matrix map whose columns are $c_1, \ldots, c_n$.

> **Check:** Compute $T(2, -1)$.
> **Answer:** $T(2, -1) = 2u_1 - u_2 = (-2 - 3,\ 6 - 3) = (-5, 3)$.

RESERVE / LOOKUP

R-Q3 · C-linear versus R-linear

LA PS6 Q3(i)–(iii)

Determine whether the following functions are (a) $\mathbb{C}$-linear and (b) $\mathbb{R}$-linear.

(i) $T:\mathbb{C}^2\to\mathbb{C}^2$, $T(x_1,x_2)=(i+x_1,\ 1+x_2)$.
(ii) $T:\mathbb{C}^4\to\mathbb{C}^2$, $T(x_1,x_2,x_3,x_4)=(x_1+ix_2,\ x_3+ix_4)$.
(iii) $T:\mathbb{C}^2\to\mathbb{C}^2$, $T(z_1,z_2)=(\mathrm{Re}(z_1),\ i\,\mathrm{Re}(z_2))$ (where $\mathrm{Re}(z)$ denotes the real part of $z\in\mathbb{C}$).

> **Say:** Over C you may multiply by any complex number; over R only by real numbers. So C-linear is the stronger condition: C-linear implies R-linear, and part (iii) shows the converse fails.

备用题，接在 L2(a)（共轭）之后最自然。板书左右两栏："$\mathbb{C}$-linear: $\lambda\in\mathbb{C}$" 和 "$\mathbb{R}$-linear: $\lambda\in\mathbb{R}$"，加法条件相同。常见错误：(iii) 用 $\lambda=i$ 的反例去说明"不是 $\mathbb{R}$-linear"。题面 (ii) 原题写成 (x_1,x_2,x_3,x_2)，已改为 x_4；官方解答 (ii) 的 "$x\in\mathbb{C}^3$" 应为 $\mathbb{C}^4$，(iii) 的 "$\mathrm{Re}(x_1+x_y)$" 应为 $\mathrm{Re}(x_1+y_1)$。

The difference. Both need $T(x+y)=T(x)+T(y)$. $\mathbb{C}$-linear needs $T(\lambda x)=\lambda T(x)$ for all $\lambda\in\mathbb{C}$; $\mathbb{R}$-linear needs it only for $\lambda\in\mathbb{R}$. Since $\mathbb{R}\subseteq\mathbb{C}$: $\mathbb{C}$-linear $\Rightarrow$ $\mathbb{R}$-linear, but not conversely.

(i) $T(0,0)=(i,1)\neq(0,0)$. Lemma 7.4(i) holds over any field, so T is neither $\mathbb{C}$-linear nor $\mathbb{R}$-linear.

(ii) Let $x=(x_1,\dots,x_4),\ y=(y_1,\dots,y_4)\in\mathbb{C}^4$ and $\lambda\in\mathbb{C}$. Then

$$\begin{aligned}T(x+y)&=\big((x_1+y_1)+i(x_2+y_2),\ (x_3+y_3)+i(x_4+y_4)\big)\\&=(x_1+ix_2,\ x_3+ix_4)+(y_1+iy_2,\ y_3+iy_4)=T(x)+T(y),\\T(\lambda x)&=(\lambda x_1+i\lambda x_2,\ \lambda x_3+i\lambda x_4)=\lambda(x_1+ix_2,\ x_3+ix_4)=\lambda T(x).\end{aligned}$$

So T is $\mathbb{C}$-linear. Real λ are a special case of complex λ, so the same computation shows T is also $\mathbb{R}$-linear. (Matrix form: $\begin{pmatrix}1&i&0&0\\0&0&1&i\end{pmatrix}$.)

(iii) Not $\mathbb{C}$-linear. Take $z=(1,0)$ and $\lambda=i$. Then $T(iz)=T(i,0)=(\mathrm{Re}(i),\ i\,\mathrm{Re}(0))=(0,0)$, but $iT(z)=i\,(\mathrm{Re}(1),\ 0)=(i,0)$. So $T(iz)\neq iT(z)$.

(iii) But $\mathbb{R}$-linear. For $a+bi,\ c+di\in\mathbb{C}$ (with $a,b,c,d\in\mathbb{R}$) and $\lambda\in\mathbb{R}$: $\mathrm{Re}\,\big((a+bi)+(c+di)\big)=a+c$, and $\mathrm{Re}\,\big(\lambda(a+bi)\big)=\mathrm{Re}(\lambda a+\lambda b\,i)=\lambda a$ because λ is real. So Re is additive and commutes with real scalars. Hence for $z=(z_1,z_2),\ w=(w_1,w_2)\in\mathbb{C}^2$ and $\lambda\in\mathbb{R}$:

$$\begin{aligned}T(z+w)&=\big(\mathrm{Re}(z_1)+\mathrm{Re}(w_1),\ i\,\mathrm{Re}(z_2)+i\,\mathrm{Re}(w_2)\big)=T(z)+T(w),\\T(\lambda z)&=\big(\lambda\,\mathrm{Re}(z_1),\ i\lambda\,\mathrm{Re}(z_2)\big)=\lambda T(z).\end{aligned}$$

So T is $\mathbb{R}$-linear but not $\mathbb{C}$-linear. The only failure is $\mathrm{Re}(iz)\neq i\,\mathrm{Re}(z)$, and i is not a real scalar.

> **Check:** Is $S:\mathbb{C}\to\mathbb{C}$, $S(z)=\overline{z}$ (complex conjugate), $\mathbb{C}$-linear? Is it $\mathbb{R}$-linear?
> **Answer:** $\mathbb{R}$-linear ($\overline{z+w}=\overline{z}+\overline{w}$, and $\overline{\lambda z}=\lambda\overline{z}$ for real λ) but not $\mathbb{C}$-linear: $S(i\cdot 1)=-i$ while $iS(1)=i$.

RESERVE / LOOKUP

R-Q4-rest · Image, kernel, rank, nullity: Q2(a), Q2(d), and Q3(iii) over R

LA PS6 Q4 for Q2(a), Q2(d) and Q3(iii)

For each linear map below, find its image, kernel, rank and nullity.

Q2(a): $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 - x_2,\ 5x_1)$.
Q2(d): $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (-x_1 + 3x_2,\ 3x_1 + 3x_2)$; this is $T(x) = (x \cdot e_1)u_1 + (x \cdot e_2)u_2$ with $u_1 = (-1, 3)$, $u_2 = (3, 3)$.
Q3(iii): $T : \mathbb{C}^2 \to \mathbb{C}^2$, $T(z_1, z_2) = (\mathrm{Re}(z_1),\ i\,\mathrm{Re}(z_2))$, as an $\mathbb{R}$-linear map (it is not $\mathbb{C}$-linear).

> **Say:** Same recipe as before: solve T(x) = 0 for the kernel and describe all outputs for the image. In Q3(iii), split each complex number into its real and imaginary parts first.

> 备用题，L8 做完有时间再做。Q3(iii) 先写 $z_1 = a + bi,\ z_2 = c + di$，得 $T = (a,\ ic)$，kernel 和 image 一眼可见。常见错误：说 $\mathrm{Im}\,T = \mathbb{C}^2$，或按 $\mathbb{C}$ 计算维数。官方解答 (3biii) 写的 "$T(0) = 0$ whenever..." 意思应是 $T(z) = 0$。

Q2(a). If $T(x_1, x_2) = 0$ then $5x_1 = 0$, so $x_1 = 0$, and $x_1 - x_2 = 0$ gives $x_2 = 0$. So $\ker T = \{0\}$, $\mathrm{nullity}\,T = 0$, and by Rank–Nullity $\mathrm{rank}\,T = 2 - 0 = 2$: $\mathrm{Im}\,T$ is a 2-dimensional subspace of $\mathbb{R}^2$, hence $\mathrm{Im}\,T = \mathbb{R}^2$. (Directly: $T\left(\frac{b}{5},\ \frac{b}{5} - a\right) = (a, b)$.)

Q2(d). $T(x) = 0$ means $-x_1 + 3x_2 = 0$ and $3x_1 + 3x_2 = 0$. The second gives $x_1 = -x_2$; substituting into the first gives $4x_2 = 0$, so $x_2 = 0 = x_1$. Hence $\ker T = \{0\}$ and $\mathrm{nullity}\,T = 0$. Since $T(x) = x_1u_1 + x_2u_2$, $\mathrm{Im}\,T = \mathrm{span}\{u_1, u_2\}$; u_1, u_2 are not multiples of each other, so they are linearly independent, hence a basis of $\mathbb{R}^2$. So $\mathrm{Im}\,T = \mathbb{R}^2$ and $\mathrm{rank}\,T = 2$. Check: $2 + 0 = 2 = \dim \mathbb{R}^2$.

Q3(iii) over $\mathbb{R}$. Write $z_1 = a + bi,\ z_2 = c + di$ with $a, b, c, d \in \mathbb{R}$. Then $T(z_1, z_2) = (a,\ ic)$.

- *Kernel:* $T(z) = 0 \iff a = c = 0$, so $\ker T = \{(bi,\ di) : b, d \in \mathbb{R}\} = \mathrm{span}_{\mathbb{R}}\{(i, 0), (0, i)\}$. These are $\mathbb{R}$-independent, so $\mathrm{nullity}_{\mathbb{R}}\,T = 2$.
- *Image:* $\mathrm{Im}\,T = \{(a,\ ic) : a, c \in \mathbb{R}\} = \mathrm{span}_{\mathbb{R}}\{(1, 0), (0, i)\}$, so $\mathrm{rank}_{\mathbb{R}}\,T = 2$.

Check: $2 + 2 = 4 = \dim_{\mathbb{R}} \mathbb{C}^2$. Note that $\mathrm{Im}\,T$ is an $\mathbb{R}$-subspace but not a $\mathbb{C}$-subspace of $\mathbb{C}^2$, consistent with T not being $\mathbb{C}$-linear.

> **Check:** Is $\mathrm{Im}\,T$ in Q3(iii) closed under multiplication by i?
> **Answer:** No: $(1, 0) \in \mathrm{Im}\,T$ but $i(1, 0) = (i, 0) \notin \mathrm{Im}\,T$, because the first entry of every output is real.

RESERVE / LOOKUP

R-Q5abc · Kernel and image: PS6 Q5(a), (b), (c)

LA PS6 Q5(a), (b), (c)

Compute the kernel and image of the following $\mathbb{R}$-linear maps:

(a) $T : \mathbb{R}^3 \to \mathbb{R}$, $T(x) = x \cdot u$, where $u = (1, 2, -1)$.
(b) $T : \mathbb{R} \to \mathbb{R}^3$, $T(x) = xu$, where $u = (1, 2, -1)$.
(c) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x - y,\ 5x)$.

> **Say:** Kernel: solve T(x) = 0. Image: take an arbitrary target and show it is hit, or describe all outputs as a span.

> 备用题，可在 L7 之前做 (a) 热身，或 L8 之后用来再核对 Rank–Nullity。常见错误：(a) 的 kernel 只说“一个平面”而不给 span；(c) 的 Im 不解方程就说“能到所有点”。

Recall (Definition 7.26): $\ker T = \{v \in V : T(v) = 0\}$ and $\operatorname{Im} T = \{w \in W : w = T(v) \text{ for some } v \in V\}$.

(a) For $x = (x_1, x_2, x_3)$, $T(x) = x_1 + 2x_2 - x_3$.

- $\ker T = \{x \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 0\}$. Solving $x_3 = x_1 + 2x_2$ gives $\ker T = \{(s, t, s + 2t) : s, t \in \mathbb{R}\} = \operatorname{span}\{(1, 0, 1), (0, 1, 2)\}$: the plane through 0 perpendicular to u.
- $\operatorname{Im} T = \mathbb{R}$: for any $r \in \mathbb{R}$, $T(r, 0, 0) = r$.

(b) $T(x) = (x, 2x, -x)$.

- $\ker T = \{0\}$: if $T(x) = 0$ then the first entry gives $x = 0$.
- $\operatorname{Im} T = \{xu : x \in \mathbb{R}\} = \operatorname{span}\{u\}$: the line through 0 in the direction $(1, 2, -1)$.

(c)

- $\ker T = \{0\}$: if $T(x, y) = (0, 0)$ then $5x = 0$, so $x = 0$, and $x - y = 0$ gives $y = 0$.
- $\operatorname{Im} T = \mathbb{R}^2$: given (a, b), solve $x - y = a$, $5x = b$ to get $x = \frac{b}{5}$, $y = \frac{b}{5} - a$; indeed $T\left(\frac{b}{5},\ \frac{b}{5} - a\right) = (a, b)$.

Rank–Nullity check: (a) $2 + 1 = 3$, (b) $0 + 1 = 1$, (c) $0 + 2 = 2$.

> **Check:** In (a), why is $\ker T$ two-dimensional without solving anything?
> **Answer:** Rank–Nullity: $\operatorname{Im} T = \mathbb{R}$ has dimension 1, so $\operatorname{nullity} T = 3 - 1 = 2$.

RESERVE / LOOKUP

R-Q6 · Injective? Surjective? A linear map from $\mathbb{R}^3$ to $\mathbb{R}^2$

LA PS6 Q6(a)–(d)

Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $f(x_1, x_2, x_3) = x_1 v_1 + x_2 v_2$ where $v_1 = (1, 0)$ and $v_2 = (1, 1)$. For each of the following, be sure to justify your answer.

(a) Show that f is an $\mathbb{R}$-linear map.
(b) Is f injective?
(c) Is f surjective?
(d) What condition on v_1, v_2 is needed to ensure that f is surjective?

> **Say:** First write f out in coordinates: f(x1, x2, x3) = (x1 + x2, x2). Notice that x3 never appears; what does that tell you about injectivity? For surjectivity, take any target (a, b) and find an input that hits it.

> 备用题，和 L5 第二个矩阵同类（$\mathbb{R}^3 \to \mathbb{R}^2$，核必非平凡）。板书先写坐标式 $(x_1 + x_2,\ x_2)$，学生一眼看出 x_3 没用到。常见错误：(c) 只说“v_1, v_2 是基所以满射”，不给具体原像 $(a - b, b, 0)$；(d) 只证一个方向。追问：由 Rank–Nullity，任何 $\mathbb{R}^3 \to \mathbb{R}^2$ 的线性映射 nullity $\geqslant 1$，所以都不单射（Corollary 7.35(ii)）。

In coordinates: $f(x_1, x_2, x_3) = x_1(1, 0) + x_2(1, 1) = (x_1 + x_2,\ x_2)$. Note that x_3 is never used.

(a) Let $x = (x_1, x_2, x_3),\ y = (y_1, y_2, y_3) \in \mathbb{R}^3$ and $\lambda \in \mathbb{R}$. Using the vector space axioms in $\mathbb{R}^2$,

$$f(x + y) = (x_1 + y_1)v_1 + (x_2 + y_2)v_2 = (x_1 v_1 + x_2 v_2) + (y_1 v_1 + y_2 v_2) = f(x) + f(y),$$
$$f(\lambda x) = (\lambda x_1)v_1 + (\lambda x_2)v_2 = \lambda(x_1 v_1 + x_2 v_2) = \lambda f(x).$$

So f is $\mathbb{R}$-linear. (Matrix form: $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$, with columns $v_1, v_2, 0$.)

(b) Not injective. $f(0, 0, 1) = 0v_1 + 0v_2 = (0, 0) = f(0, 0, 0)$, but $(0, 0, 1) \neq (0, 0, 0)$. (In fact $\ker f = \operatorname{span}\{e_3\} \neq \{0\}$.)

(c) Surjective. Let $(a, b) \in \mathbb{R}^2$ be arbitrary. Then $f(a - b,\ b,\ 0) = (a - b)(1, 0) + b(1, 1) = (a - b + b,\ b) = (a, b)$. So every point of $\mathbb{R}^2$ is an output of f.

(d) For general v_1, v_2, $\operatorname{Im} f = \{x_1 v_1 + x_2 v_2 : x_1, x_2 \in \mathbb{R}\} = \operatorname{span}\{v_1, v_2\}$. So f is surjective if and only if v_1, v_2 span $\mathbb{R}^2$; for two vectors in $\mathbb{R}^2$ this happens exactly when v_1, v_2 are linearly independent, i.e. form a basis of $\mathbb{R}^2$.

- If v_1, v_2 span $\mathbb{R}^2$: any (a, b) equals $\alpha v_1 + \beta v_2$ for some $\alpha, \beta \in \mathbb{R}$, and then $f(\alpha, \beta, 0) = (a, b)$.
- If v_1, v_2 are linearly dependent, say $v_2 = k v_1$ (or $v_1 = k v_2$), then $\operatorname{Im} f = \operatorname{span}\{v_1\}$ (or $\operatorname{span}\{v_2\}$), which has dimension at most $1 < 2$. So $\operatorname{Im} f \neq \mathbb{R}^2$ and f is not surjective.

> **Check:** If instead $v_1 = (1, 2)$ and $v_2 = (2, 4)$, what is $\operatorname{Im} f$? Is f surjective?
> **Answer:** $v_2 = 2v_1$, so $\operatorname{Im} f = \operatorname{span}\{(1, 2)\}$, the line $y = 2x$. Not surjective: e.g. $(1, 0) \notin \operatorname{Im} f$.

RESERVE / LOOKUP

R-Q9 · The reflection h: PS6 Q9(a)–(d)

LA PS6 Q9(a)–(d)

Let $h : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map given by $h(x) = Ax$ where $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

(a) Does the information above enable us to find, uniquely, $h(x)$ for all $x \in \mathbb{R}^2$? Justify your answer.
(b) Write h in standard basis form.
(c) Write h in algebraic form.
(d) What transformation of $\mathbb{R}^2$ does h represent?

> **Say:** The columns of A tell you where e1 and e2 go. Swapping e1 and e2 is a reflection: which line stays fixed?

> 备用题，是 L6（Q9(e)）的前半部分；学生对 h 不熟时先口头过一遍。常见错误：(a) 只说“是”不给理由。

(a) Yes. For each $x \in \mathbb{R}^2$, Ax is one specific vector, so $h(x)$ is determined. Alternatively: h is linear (Fact 7.14(i)) and we know $h(e_1), h(e_2)$, so Lemma 7.9 determines h.

(b) By Fact 7.14(ii), $h(e_i)$ is the ith column of A: $h(e_1) = (0,1) = e_2$ and $h(e_2) = (1,0) = e_1$. So $h : e_1 \mapsto e_2,\ e_2 \mapsto e_1$.

(c) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$ for all $x, y \in \mathbb{R}$, so $h(x, y) = (y, x)$.

(d) h fixes every point of the line $y = x$ ($h(t,t) = (t,t)$) and sends $(1,-1)$ to $-(1,-1)$. So h is the reflection in the line $y = x$.

> **Check:** What is $h \circ h$?
> **Answer:** The identity: $h(h(x,y)) = h(y,x) = (x,y)$, and $A^2 = I$. Reflecting twice undoes the reflection.

RESERVE / LOOKUP

R-Q10 · Rotations and the addition formulas

LA PS6 Q10(a)–(c)

Let $f_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ be the transformation that rotates anti-clockwise by θ radians. (It is helpful to know that the point given by rotating e_1 by θ radians is $(\cos\theta, \sin\theta)$.)

(a) Find each of the following, in whichever order you prefer: (i) the algebraic form of f_θ; (ii) the matrix form of f_θ; (iii) the standard basis form of f_θ.

(b) Note, given $\theta, \varphi \in \mathbb{R}$, that $f_\theta \circ f_\varphi = f_\varphi \circ f_\theta = f_{\theta+\varphi}$.
(i) Using (a)(ii), write out the matrix form of $f_{\theta+\varphi}$.
(ii) Using the matrix forms of f_θ and f_φ, find the matrix form of $f_\theta \circ f_\varphi$.
(iii) Use (i) and (ii) to find identities relating to $\sin(\theta+\varphi)$ and $\cos(\theta+\varphi)$.

(c) Observe that $f_\theta^{-1} = f_{-\theta}$. By comparing the matrix form of $f_{-\theta}$ and the inverse of the matrix form of f_θ, deduce one property of $\cos$ and one of $\sin$.

> **Say:** First find where e1 and e2 go; that gives the matrix straight away. Then the trick in (b): the same map has two matrices written in two different ways, so their entries must be equal, and out pop the addition formulas.

> 备用题，适合学得快的学生。板书画单位圆：$e_1 \to (\cos\theta, \sin\theta)$，$e_2 \to (-\sin\theta, \cos\theta)$。关键论证：同一个线性映射只有一个矩阵（列 $= e_i$ 的像），所以可以比较矩阵元素。常见错误：R_θ 中 $-\sin\theta$ 的位置放错。

We use that rotation about the origin is linear (it carries the parallelogram for $v + w$ to the parallelogram for $f_\theta(v) + f_\theta(w)$, and commutes with scaling), so by Lemma 7.9 it is determined by $f_\theta(e_1), f_\theta(e_2)$.

(a)(iii) $f_\theta(e_1) = (\cos\theta, \sin\theta)$ (given). The vector e_2 is e_1 turned by $\pi/2$, i.e. $e_2 = g(e_1)$ with $g = f_{\pi/2}$, $g(x, y) = (-y, x)$ from Q8. Rotations commute, so $f_\theta(e_2) = f_\theta(g(e_1)) = g(f_\theta(e_1)) = g(\cos\theta, \sin\theta) = (-\sin\theta, \cos\theta)$. Standard basis form: $f_\theta : e_1 \mapsto (\cos\theta, \sin\theta),\ e_2 \mapsto (-\sin\theta, \cos\theta)$.

(a)(ii) Columns from (iii): $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

(a)(i) $R_\theta \begin{pmatrix} x \\ y \end{pmatrix}$ gives $f_\theta(x, y) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta)$.

(b)(i) $R_{\theta+\varphi} = \begin{pmatrix} \cos(\theta+\varphi) & -\sin(\theta+\varphi) \\ \sin(\theta+\varphi) & \cos(\theta+\varphi) \end{pmatrix}$.

(b)(ii) The matrix of $f_\theta \circ f_\varphi$ is $R_\theta R_\varphi$:

$$R_\theta R_\varphi = \begin{pmatrix} \cos\theta\cos\varphi - \sin\theta\sin\varphi & -(\sin\theta\cos\varphi + \cos\theta\sin\varphi) \\ \sin\theta\cos\varphi + \cos\theta\sin\varphi & \cos\theta\cos\varphi - \sin\theta\sin\varphi \end{pmatrix}.$$

(b)(iii) Both matrices represent the same linear map $f_{\theta+\varphi} = f_\theta \circ f_\varphi$. A linear map has only one matrix (its columns must be the images of e_1, e_2, Fact 7.14(ii)), so the entries agree:

$$\cos(\theta+\varphi) = \cos\theta\cos\varphi - \sin\theta\sin\varphi, \qquad \sin(\theta+\varphi) = \sin\theta\cos\varphi + \cos\theta\sin\varphi.$$

(c) $\det R_\theta = \cos^2\theta + \sin^2\theta = 1$, so $R_\theta^{-1} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$. By (a)(ii) the matrix of $f_{-\theta}$ is $\begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix}$. Since $f_{-\theta} = f_\theta^{-1}$, the two matrices are equal, so $\cos(-\theta) = \cos\theta$ ($\cos$ is even) and $\sin(-\theta) = -\sin\theta$ ($\sin$ is odd).

> **Check:** Which θ gives the map g from Q8, and which gives $f(x,y)=(-x,-y)$ from Q7?
> **Answer:** $\theta=\pi/2$ gives $\begin{pmatrix}0&-1\\1&0\end{pmatrix}=A_g$; $\theta=\pi$ gives $-I=A_f$.

RESERVE / LOOKUP

R-Q11 · Additive implies Q-linear

LA PS6 Q11

Let $m,n\in\mathbb{N}$ and suppose that $f:\mathbb{Q}^n\to\mathbb{Q}^m$ satisfies $f(x+y)=f(x)+f(y)$ for all $x,y\in\mathbb{Q}^n$. Prove that f is a $\mathbb{Q}$-linear map.

> **Say:** We are only given the addition rule, and we need scalar multiplication for every rational number. Build it up: first 0, then minus signs, then whole numbers, then one over q, then p over q.

> 备用题，给学得快的学生。板书按 $0\to-1\to p\in\mathbb{N}\to 1/q\to p/q$ 的顺序写五步。常见错误：Step 4 直接写 $f(x/q)=f(x)/q$，没有说明是对向量 x/q 用 Step 3。追问：为什么推不到 $\sqrt{2}$?

Additivity is given, so we must show $f(\lambda x)=\lambda f(x)$ for all $x\in\mathbb{Q}^n$ and all $\lambda\in\mathbb{Q}$. We build up from whole numbers to fractions.

Step 1 ($\lambda=0$). $f(0)=f(0+0)=f(0)+f(0)$; subtracting $f(0)$ gives $f(0)=0$. So $f(0\cdot x)=f(0)=0=0\cdot f(x)$.

Step 2 (negatives). $0=f(0)=f(x+(-x))=f(x)+f(-x)$, so $f(-x)=-f(x)$.

Step 3 ($\lambda=p\in\mathbb{N}$). By induction on p: $f(1\cdot x)=1\cdot f(x)$, and if $f(px)=pf(x)$ then $f((p+1)x)=f(px+x)=f(px)+f(x)=pf(x)+f(x)=(p+1)f(x)$.

Step 4 ($\lambda=1/q$, $q\in\mathbb{N}$). Apply Step 3 with $p=q$ to the vector $\frac{1}{q}x\in\mathbb{Q}^n$: $f(x)=f\big(q\cdot\frac{1}{q}x\big)=q\,f\big(\frac{1}{q}x\big)$, so $f\big(\frac{1}{q}x\big)=\frac{1}{q}f(x)$.

Step 5 (any $\lambda\in\mathbb{Q}$). Every nonzero rational is $\frac{p}{q}$ or $-\frac{p}{q}$ with $p,q\in\mathbb{N}$. By Steps 3 and 4, $f\big(\frac{p}{q}x\big)=f\big(p\cdot\frac{1}{q}x\big)=p\,f\big(\frac{1}{q}x\big)=\frac{p}{q}f(x)$, and by Step 2, $f\big(-\frac{p}{q}x\big)=-f\big(\frac{p}{q}x\big)=-\frac{p}{q}f(x)$. Together with Step 1, $f(\lambda x)=\lambda f(x)$ for all $\lambda\in\mathbb{Q}$.

Hence f satisfies both conditions of Definition 7.1 with $\mathbb{K}=\mathbb{Q}$, so f is $\mathbb{Q}$-linear.

Remark. The argument only ever reaches rational scalars. For maps $\mathbb{R}^n\to\mathbb{R}^m$, additivity alone does not give $f(\sqrt{2}\,x)=\sqrt{2}\,f(x)$; one needs extra input, such as continuity.

> **Check:** Why does this argument not show $f(\sqrt{2}\,x)=\sqrt{2}\,f(x)$ for an additive map $f:\mathbb{R}^n\to\mathbb{R}^m$?
> **Answer:** Additivity only produces rational multiples $\frac{p}{q}$, and $\sqrt{2}$ is not of that form, so none of the steps reaches it.

RESERVE / LOOKUP

R-Q12 · RREF: rank and the dimension of the solution space

LA PS6 Q12(a)(b)

Assume $A \in M_{m,n}(\mathbb{R})$ is in reduced row echelon form with k leading 1's, and let $T_A : \mathbb{R}^n \to \mathbb{R}^m$, $x \mapsto Ax$.

(a) Show that $e_1, e_2, \ldots, e_k$ are a basis of $\operatorname{Im} T_A$ and compute $\operatorname{rank} T_A$.
(b) Now prove that $\dim S(A, 0) = n - k$, possibly by relying on a result from lectures.

> **Say:** The image of a matrix map is the span of its columns. In reduced row echelon form, the pivot columns are exactly e1 up to ek, and every other column lives in their span.

> 备用题，(b) 直接用 Rank–Nullity（Theorem 7.31）。先写 $\operatorname{Im} T_A = \operatorname{span}\{a_1, \ldots, a_n\}$，再证两个方向的包含。官方解答把 V_k 写在 $\mathbb{R}^n$ 中、条件为 $x_{k+1} = \cdots = x_n = 0$，应为 $\mathbb{R}^m$ 和 y_m。常见错误：只证一个方向；把这里的 e_i 当成 $\mathbb{R}^n$ 的向量。

Let $a_1, \ldots, a_n \in \mathbb{R}^m$ be the columns of A. By Fact 7.14, $T_A(x) = Ax = x_1 a_1 + \cdots + x_n a_n$, so $\operatorname{Im} T_A = \operatorname{span}\{a_1, \ldots, a_n\}$.

(a) Let $V_k = \{y \in \mathbb{R}^m : y_{k+1} = \cdots = y_m = 0\} = \operatorname{span}\{e_1, \ldots, e_k\}$, where $e_i \in \mathbb{R}^m$.

- $\operatorname{Im} T_A \subseteq V_k$*:* in RREF with k leading 1's, the leading 1's sit in rows $1, \ldots, k$, and rows $k+1, \ldots, m$ are zero. So every column a_j has zero entries in positions $k+1, \ldots, m$, i.e. $a_j \in V_k$. As V_k is a subspace, $\operatorname{span}\{a_1, \ldots, a_n\} \subseteq V_k$.
- $V_k \subseteq \operatorname{Im} T_A$*:* in *reduced* row echelon form, the column containing the ith leading 1 has a 1 in row i and 0 everywhere else, so it equals e_i. Thus each of $e_1, \ldots, e_k$ is a column of A and lies in $\operatorname{Im} T_A$; since $\operatorname{Im} T_A$ is a subspace (Lemma 7.27), $\operatorname{span}\{e_1, \ldots, e_k\} \subseteq \operatorname{Im} T_A$.

Hence $\operatorname{Im} T_A = \operatorname{span}\{e_1, \ldots, e_k\}$. The vectors $e_1, \ldots, e_k$ are linearly independent, so they form a basis of $\operatorname{Im} T_A$, and $\operatorname{rank} T_A = k$.

(b) $S(A, 0) = \{x \in \mathbb{R}^n : Ax = 0\} = \ker T_A$. By the Rank–Nullity Theorem (Theorem 7.31) for $T_A : \mathbb{R}^n \to \mathbb{R}^m$,

$$\dim S(A, 0) = \operatorname{nullity} T_A = n - \operatorname{rank} T_A = n - k.$$

(Without Rank–Nullity: the $n - k$ non-pivot variables are free, and the LA1a method gives one basis vector of $S(A, 0)$ for each free variable.)

Correction to the official solution: the set V_k lives in $\mathbb{R}^m$, with conditions $y_{k+1} = \cdots = y_m = 0$, not in $\mathbb{R}^n$.

> **Check:** For $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, what are $\operatorname{rank} T_A$ and $\operatorname{nullity} T_A$?
> **Answer:** $k = 2$ leading 1's, so rank 2 and nullity $3 - 2 = 1$; the kernel is $\operatorname{span}\{(-2, 1, 0)\}$.

RESERVE / LOOKUP

R-Q13 · Independent columns if and only if nullity 0

LA PS6 Q13

Let $A \in M_{m,n}(\mathbb{R})$ and let $T_A : \mathbb{R}^n \to \mathbb{R}^m$ be the corresponding linear map $T_A(x) = Ax$. Show that the column vectors of A, $a_1, \ldots, a_n$, are linearly independent if and only if $\operatorname{nullity} T_A = 0$.

> **Say:** Ax is just x1 times the first column plus and so on up to xn times the last column. So 'Ax = 0 only for x = 0' is, word for word, the definition of the columns being independent.

> 备用题。关键一行：$Ax = x_1a_1 + \cdots + x_na_n$（Fact 7.14 / Example 7.30），写出来以后两个方向几乎就是定义。常见错误：把 nullity 0 理解成 $\ker T_A = \varnothing$（核总含 0）。

By Fact 7.14, $T_A(x) = Ax = x_1a_1 + \cdots + x_na_n$. So

$$x \in \ker T_A \iff x_1a_1 + \cdots + x_na_n = 0. \qquad (*)$$

Also $\operatorname{nullity} T_A = \dim \ker T_A = 0$ if and only if $\ker T_A = \{0\}$, since $\{0\}$ is the only subspace of dimension 0.

($\Rightarrow$) Suppose $a_1, \ldots, a_n$ are linearly independent. If $x \in \ker T_A$, then by $(*)$ $x_1a_1 + \cdots + x_na_n = 0$, so every $x_i = 0$, i.e. $x = 0$. Hence $\ker T_A = \{0\}$ and $\operatorname{nullity} T_A = 0$.

($\Leftarrow$) Suppose $\operatorname{nullity} T_A = 0$, so $\ker T_A = \{0\}$. If $x_1a_1 + \cdots + x_na_n = 0$ for some scalars x_i, then by $(*)$ the vector $x = (x_1, \ldots, x_n)$ lies in $\ker T_A = \{0\}$, so $x_1 = \cdots = x_n = 0$. Hence $a_1, \ldots, a_n$ are linearly independent.

> **Check:** Can a 2×3 real matrix A have $\operatorname{nullity} T_A = 0$?
> **Answer:** No: three vectors in $\mathbb{R}^2$ are always linearly dependent, so by this result $\operatorname{nullity} T_A \geqslant 1$.

RESERVE / LOOKUP

R-Q14 · Composition of bijective linear maps and its inverse

LA PS6 Q14

Let V be a vector space over $\mathbb{K}$. Assume $T : V \to V$ and $S : V \to V$ are bijective linear maps, i.e., $\ker T = \ker S = \{0\}$ and $\operatorname{Im} T = \operatorname{Im} S = V$. Show that $R := S \circ T$ is bijective and that $R^{-1} = T^{-1} \circ S^{-1}$.

> **Say:** To undo 'first T, then S', you undo S first and then T, like taking off your shoes before your socks. For bijectivity, chase an element through: the kernel for injective, a preimage for surjective.

> 备用题。板书分三块：kernel、image、inverse。常见错误：把逆写成 $S^{-1} \circ T^{-1}$（顺序反了）。追问：用 L6 的 g, h，$(h \circ g)^{-1}$ 是什么?

R is linear: $R(v+w) = S(T(v)+T(w)) = S(T(v)) + S(T(w)) = R(v) + R(w)$, and $R(\lambda v) = S(\lambda T(v)) = \lambda S(T(v)) = \lambda R(v)$.

$\ker R = \{0\}$: let $x \in \ker R$. Then $S(T(x)) = 0$, so $T(x) \in \ker S = \{0\}$, i.e. $T(x) = 0$. So $x \in \ker T = \{0\}$, i.e. $x = 0$. Hence R is injective: if $R(x) = R(y)$ then $R(x-y) = 0$, so $x - y = 0$.

$\operatorname{Im} R = V$: let $w \in V$. Since $\operatorname{Im} S = V$, there is $u \in V$ with $S(u) = w$; since $\operatorname{Im} T = V$, there is $v \in V$ with $T(v) = u$. Then $R(v) = S(T(v)) = S(u) = w$. So R is surjective, and therefore bijective.

Inverse: using associativity of composition,

$$(T^{-1} \circ S^{-1}) \circ (S \circ T) = T^{-1} \circ (S^{-1} \circ S) \circ T = T^{-1} \circ T = \mathrm{id}_V,$$

$$(S \circ T) \circ (T^{-1} \circ S^{-1}) = S \circ (T \circ T^{-1}) \circ S^{-1} = S \circ S^{-1} = \mathrm{id}_V.$$

So $T^{-1} \circ S^{-1}$ is a two-sided inverse of R, i.e. $R^{-1} = T^{-1} \circ S^{-1}$ (“socks and shoes”: undo the last step first).

> **Check:** With g (rotation by $\pi/2$) and h (reflection in $y = x$) from Q8/Q9, what is $(h \circ g)^{-1}$?
> **Answer:** $(h \circ g)^{-1} = g^{-1} \circ h^{-1} = g^{-1} \circ h$. Since $h \circ g$ is the reflection $(x, y) \mapsto (x, -y)$, it is its own inverse.

RESERVE / LOOKUP

R-Q15 · Nullity of a composition: two explicit examples

LA PS6 Q15(a)(b)

Find linear maps $T : \mathbb{R}^3 \to \mathbb{R}^3$, $S : \mathbb{R}^3 \to \mathbb{R}^3$ so that

(a) $\operatorname{nullity}(S \circ T) = 2$, $\operatorname{nullity} S = 1$, $\operatorname{nullity} T = 1$ (note then $\operatorname{nullity}(S \circ T) = \operatorname{nullity} S + \operatorname{nullity} T$);
(b) $\operatorname{nullity}(S \circ T) = 2$, $\operatorname{nullity} S = 2$, $\operatorname{nullity} T = 1$ (note then $\operatorname{nullity}(S \circ T) < \operatorname{nullity} S + \operatorname{nullity} T$).

> **Say:** Start with the simplest maps you know: ones that kill a coordinate. A vector is killed by S∘T exactly when T sends it into the kernel of S.

> 备用题。用“把某个坐标变成 0”的对角映射最快。常见错误：只给出 S, T 而不验证三个 nullity；或以为总有 $\operatorname{nullity}(S \circ T) = \operatorname{nullity} S + \operatorname{nullity} T$。追问：$\operatorname{nullity}(S \circ T)$ 能小于 $\operatorname{nullity} T$ 吗？（不能，$\ker T \subseteq \ker(S \circ T)$）

(a) Let $T(x, y, z) = (0, y, z)$ and $S(x, y, z) = (x, 0, z)$. Both are linear (matrices $\operatorname{diag}(0, 1, 1)$ and $\operatorname{diag}(1, 0, 1)$).

- $T(x, y, z) = 0 \iff y = z = 0$, so $\ker T = \{(x, 0, 0) : x \in \mathbb{R}\} = \operatorname{span}\{e_1\}$ and $\operatorname{nullity} T = 1$.
- $S(x, y, z) = 0 \iff x = z = 0$, so $\ker S = \operatorname{span}\{e_2\}$ and $\operatorname{nullity} S = 1$.
- $(S \circ T)(x, y, z) = S(0, y, z) = (0, 0, z)$, which is 0 exactly when $z = 0$. So $\ker(S \circ T) = \operatorname{span}\{e_1, e_2\}$ and $\operatorname{nullity}(S \circ T) = 2 = 1 + 1$.

(b) Keep $T(x, y, z) = (0, y, z)$ and take $S(x, y, z) = (0, 0, z)$.

- $\ker S = \{(x, y, 0) : x, y \in \mathbb{R}\} = \operatorname{span}\{e_1, e_2\}$, so $\operatorname{nullity} S = 2$; $\operatorname{nullity} T = 1$ as before.
- $(S \circ T)(x, y, z) = S(0, y, z) = (0, 0, z)$, so $\ker(S \circ T) = \operatorname{span}\{e_1, e_2\}$ and $\operatorname{nullity}(S \circ T) = 2 < 2 + 1$.

Why the difference. $x \in \ker(S \circ T)$ means $T(x) \in \ker S$. In (a), $\ker S = \operatorname{span}\{e_2\}$ lies inside $\operatorname{Im} T = \operatorname{span}\{e_2, e_3\}$, so all of it is reached by T. In (b), only the e_2-direction of $\ker S$ is reached; the e_1-direction is never an output of T, so it adds nothing to $\ker(S \circ T)$.

> **Check:** Could $\operatorname{nullity}(S \circ T)$ be 0 when $\operatorname{nullity} T = 1$?
> **Answer:** No: if $T(x) = 0$ then $S(T(x)) = S(0) = 0$, so $\ker T \subseteq \ker(S \circ T)$ and $\operatorname{nullity}(S \circ T) \geqslant 1$.

Sources and scope

Pure: Problem Sheet 6 (ex6.pdf), its Part A and full solutions, and the Part II lecture summary (Week 7 lectures of 2 and 3 November), downloaded from the MATH10027 Blackboard course on 4 November 2026.

Linear: Linear Algebra Problem Sheet 6 and its solutions, and Chapter 7 lecture notes parts 1–3 with the annotated Lectures 7.1 and 7.2, downloaded from the MATH10015 Blackboard course on 4 November 2026.

Every solution has been re-derived and checked against the official solutions; obvious typographical errors in the sources are corrected. Hand-in questions (Pure PS6 Assigned Question, due 13:00 Thursday 12 November; Linear PS6 Q1, due noon Wednesday 11 November) are hints only. The student worksheets contain questions and the key-ideas page only — no answers.

> 本页给老师看（含答案），不要发给学生。学生版只用 Student PDF。